

Bayesian Deep Learning for Uncertainty Quantification: A Survey

Bayesian deep learning aims to make neural networks useful not only as predictors but also as probabilistic systems that represent uncertainty. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around Bayesian neural networks, posterior inference, variational and Monte Carlo approximations, dropout, ensembles, calibration, epistemic and aleatoric uncertainty, out-of-distribution detection, active learning, Bayesian optimization, probabilistic architectures, applications, and evaluation. The central argument is that uncertainty quantification is a full-stack problem: priors, approximations, training objectives, predictive distributions, calibration metrics, data shift, decision costs, and user-facing risk controls must be designed together.

Keywords:

Bayesian deep learning; uncertainty quantification; Bayesian neural networks; calibration; epistemic uncertainty; out-of-distribution detection

1 Introduction

Deep neural networks often make accurate predictions while hiding the limits of their knowledge. In safety-critical, scientific, medical, financial, robotic, and autonomous systems, a point estimate is rarely enough: users need to know when a model is uncertain, whether that uncertainty comes from reducible ignorance or irreducible noise, and how much a decision should trust the prediction. Bayesian deep learning addresses this need by placing probability distributions over parameters, functions, latent variables, predictions, or decision outcomes [16, 58, 70, 144, 174, 185, 199, 265].

The field is broad because exact Bayesian inference in modern neural networks is usually intractable. Researchers therefore use variational inference, Monte Carlo dropout, Laplace approximations, stochastic-gradient MCMC, ensembles, Bayesian last layers, deep Gaussian processes, neural processes, evidential objectives, conformal wrappers, and calibration procedures. These methods differ in what uncertainty they represent, how much computation they require, and how faithfully they approximate a posterior [16, 58, 70, 144, 174, 185, 199, 227].

This survey treats Bayesian deep learning for uncertainty quantification as a design problem that connects modeling, inference, evaluation, and deployment. A method is useful only if its uncertainty estimates remain meaningful under dataset shift, class imbalance, limited labels, long-tailed events, adversarial perturbations, and downstream decisions.

2 Background

2.1 Uncertainty Types

Uncertainty in deep learning is commonly separated into epistemic and aleatoric components. Epistemic uncertainty reflects what the model does not know because data are limited or distributional coverage is poor; it should decrease with relevant data. Aleatoric uncertainty reflects noise, ambiguity, or inherent randomness in observations and labels; it may persist even with more data. Practical systems also face model misspecification, dataset shift, annotation ambiguity, and decision uncertainty [16, 22, 34, 42, 70, 184, 199, 273].

2.2 Bayesian Neural Networks

Bayesian neural networks place distributions over weights or functions and infer a posterior conditioned on data. This idea is conceptually attractive because predictive distributions follow from integrating over plausible models. The difficulty is scale: modern networks have millions or billions of parameters, complicated likelihoods, and nonconvex posteriors. Approximate inference is therefore not a minor implementation detail; it largely determines the behavior of the uncertainty estimate [16, 58, 70, 144, 174, 185, 199, 227].

2.3 Calibration and Decision Usefulness

An uncertainty estimate should be calibrated: events assigned probability p should occur about p of the time under appropriate evaluation conditions. Calibration alone is not sufficient, because a forecast can be calibrated yet too diffuse or poorly ranked. Useful uncertainty should also support abstention, active labeling, exploration, risk-aware optimization, human review, and safe fallback behavior [16, 70, 174, 184, 185, 227, 229, 265].

3 Methods

3.1 Search Protocol and Corpus Construction

The literature corpus for this survey was generated during this run from the arXiv structured API. Queries combined Bayesian deep learning, Bayesian neural networks, uncertainty quantification, posterior inference, variational inference, Monte Carlo dropout, deep ensembles, Laplace approximations, stochastic-gradient MCMC, calibration, out-of-distribution detection, active learning, Bayesian optimization, probabilistic forecasting, neural processes, evidential learning, conformal prediction, and uncertainty-aware applications. Records were retained when the title or abstract explicitly connected Bayesian or approximate Bayesian neural methods with uncertainty quantification, calibration, distribution shift, probabilistic prediction, or uncertainty-aware decisions. The final bibliography cites 300 unique arXiv papers.

3.2 Taxonomy

We organize the literature along eight axes. The first is *Bayesian object*: weights, functions, features,

latent variables, outputs, or decisions. The second is *inference approximation*: variational families, dropout, Laplace approximations, ensembles, stochastic-gradient MCMC, Bayesian last layers, or hybrid methods. The third is *uncertainty type*: epistemic, aleatoric, predictive, distributional, or decision uncertainty. The fourth is *architecture*: convolutional networks, transformers, graph neural networks, recurrent models, generative models, Gaussian-process hybrids, or neural processes. The fifth is *evaluation*: calibration, negative log likelihood, proper scoring rules, OOD detection, selective risk, robustness, and decision utility. The sixth is *data regime*: small data, long-tailed labels, noisy labels, domain shift, active learning, or continual updates. The seventh is *application*: vision, medical imaging, forecasting, robotics, reinforcement learning, scientific modeling, or language. The eighth is *deployment*: compute budget, latency, interpretability, monitoring, and human oversight.

3.3 Posterior Inference for Bayesian Neural Networks

Posterior inference methods define how probability enters a neural model. Mean-field variational methods are scalable but can underestimate posterior correlations. Richer variational families and flows improve expressivity but add optimization complexity. Laplace approximations exploit local curvature; stochastic-gradient MCMC aims for samples but can be expensive; Bayesian last layers and neural-linear models trade depth uncertainty for tractability [16, 58, 70, 144, 174, 185, 199, 227].

3.4 Variational Inference, Dropout, and Scalable Approximations

Scalable approximations made Bayesian deep learning practical. Variational inference converts posterior inference into optimization, while dropout-based methods reinterpret stochastic subnetworks as approximate Bayesian predictions. These approaches are attractive because they fit existing training pipelines, but their uncertainty quality depends on the variational family, prior choice, objective, data scale, and mismatch between training and evaluation [16, 58, 70, 144, 185, 199, 227, 265].

3.5 Ensembles, Approximate Bayesian Methods, and Calibration

Deep ensembles are often strong empirical baselines for uncertainty and calibration. They combine different initial conditions, data orderings, or model vari-

ants to approximate diversity in function space. Calibration methods such as temperature scaling can correct confidence after training, but they may not fix OOD uncertainty or poor epistemic coverage. Strong evaluations should compare Bayesian methods with ensembles, calibration baselines, and proper scoring rules [16, 70, 174, 184, 185, 227, 229, 265].

3.6 Epistemic and Aleatoric Predictive Uncertainty

Many applications need the model to separate reducible and irreducible uncertainty. Heteroscedastic likelihoods, distributional heads, mutual-information criteria, variance decompositions, and Bayesian predictive distributions are used for this purpose. The distinction matters operationally: epistemic uncertainty can trigger data acquisition or human review, whereas aleatoric uncertainty may call for risk-aware decisions or wider prediction intervals [16, 22, 34, 42, 70, 184, 199, 273].

3.7 Out-of-Distribution Detection, Robustness, and Shift

Uncertainty estimates are most valuable when the test distribution differs from training. Bayesian and approximate Bayesian methods are used for OOD detection, anomaly detection, selective classification, abstention, robust prediction, and uncertainty under adversarial or domain shift. The hard problem is that high confidence can persist far from the training distribution, especially when the model learns brittle shortcuts [34, 42, 55, 70, 174, 199, 227, 265].

3.8 Active Learning, Bayesian Optimization, and Decisions

Uncertainty is not only a diagnostic signal; it guides action. Active learning chooses examples expected to reduce uncertainty or improve decision boundaries. Bayesian optimization uses surrogate uncertainty to trade off exploration and exploitation. Reinforcement learning and risk-sensitive decision making use uncertainty for exploration, safety, and conservative control. These settings test whether uncertainty estimates are useful under sequential costs [16, 104, 141, 144, 145, 265, 266, 295].

3.9 Probabilistic Architectures and Generative Models

Bayesian deep learning overlaps with probabilistic deep learning. Variational autoencoders, normalizing flows, deep Gaussian processes, neural processes,

latent-variable sequence models, diffusion-style predictive distributions, and probabilistic forecasting networks provide alternatives to weight-space posteriors. These methods often model output or function uncertainty directly, which can be more practical than full Bayesian treatment of all weights [22, 42, 58, 144, 174, 185, 199, 273].

3.10 Applications, Benchmarks, and Evaluation Protocols

Applications in medical imaging, autonomous driving, robotics, computer vision, scientific modeling, natural language processing, and forecasting use uncertainty to support triage, human review, data acquisition, safety margins, and model monitoring. Evaluation should include calibration under shift, OOD detection, selective risk, proper scoring rules, label noise, subgroup behavior, and decision impact rather than only accuracy [55, 72, 123, 183–185, 273, 300].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, showing how work spans Bayesian neural networks, variational approximations, ensembles, calibration, epistemic and aleatoric uncertainty, OOD detection, active learning, probabilistic architectures, applications, and evaluation.

Bayesian neural networks and posterior inference

The Bayesian neural networks and posterior inference cluster contains work that develops methods, systems, datasets, or analyses central to this survey [9, 58, 111, 123, 144, 145, 174, 229, 300]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [18, 104, 118, 178, 194, 213, 269, 286, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 37, 47, 75, 79, 86, 101, 204, 235]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [63, 87, 106, 117, 142, 176, 236, 260, 267]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [28, 44, 48, 138, 156, 209, 243, 258, 271]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [97, 143, 153, 196, 205, 218, 222, 256, 289]. Additional retained studies broaden the cluster with alternative architec-

tures, training signals, data sources, and deployment assumptions [61, 107, 126, 137, 163, 197, 216, 228, 251]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [25, 41, 46, 114, 129, 148, 173, 189, 250]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [29, 60, 108, 125, 146, 200, 203, 219, 249]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [8, 81, 105, 135, 140, 254, 277, 296, 299]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [23, 67, 131, 175, 187, 212, 241, 242, 291]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 21, 26, 89, 93, 167, 192, 288, 294]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [65, 119, 207, 268].

Variational inference and scalable approximations

The variational inference and scalable approximations cluster contains work that develops methods, systems, datasets, or analyses central to this survey [16, 22, 77, 90, 133, 136, 181, 183, 273]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [5, 76, 78, 85, 95, 127, 130, 276, 292]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [45, 92, 100, 103, 158, 220, 248, 264, 283]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [15, 20, 102, 115, 168, 188, 195, 230, 261]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [4, 14, 57, 62, 120, 169, 170, 252, 255].

Ensembles, approximate bayesian methods, and calibration

The ensembles, approximate Bayesian methods, and calibration cluster contains work that develops methods, systems, datasets, or analyses central to this survey [34, 42, 55, 70, 157, 184, 185, 227, 265]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [43, 50, 69, 84, 109, 112, 150, 186, 221]. Additional retained studies broaden the cluster with alternative architectures, training

signals, data sources, and deployment assumptions [88, 91, 128, 147, 151, 152, 179, 201, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 19, 177, 214, 215, 237, 246, 253, 259]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [27, 36, 52, 80, 164, 202, 247, 270, 274]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [7, 53, 68, 73, 166, 206, 238, 280, 287]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [39, 71, 94, 134, 162, 211, 225, 263, 278]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [282].

Epistemic and aleatoric uncertainty in prediction

The epistemic and aleatoric uncertainty in prediction cluster contains work that develops methods, systems, datasets, or analyses central to this survey [31, 161, 198, 199, 224, 245, 266, 284, 285]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [35, 51, 64, 74, 82, 223, 262, 281, 297]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 59, 98, 99, 132, 226, 231].

Out-of-distribution detection, robustness, and shift

The out-of-distribution detection, robustness, and shift cluster contains work that develops methods, systems, datasets, or analyses central to this survey [110, 124, 155, 172].

Active learning, bayesian optimization, and decision making

The active learning, Bayesian optimization, and decision making cluster contains work that develops methods, systems, datasets, or analyses central to this survey [83, 113, 141, 193, 279, 295].

Probabilistic deep learning architectures and generative models

The probabilistic deep learning architectures and generative models cluster contains work that develops methods, systems, datasets, or analyses central to this survey [11, 40, 49, 275].

Applications, benchmarks, and evaluation protocols

The applications, benchmarks, and evaluation protocols cluster contains work that develops methods,

systems, datasets, or analyses central to this survey [2, 13, 54, 72, 160, 180, 233, 234, 239]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [17, 24, 30, 32, 122, 208, 217, 240, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [33, 56, 66, 139, 149, 171, 190, 210, 232]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 38, 96, 116, 121, 165, 191, 244, 257]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [154, 159, 182, 272].

4 Challenges and Prospects

4.1 Approximation Quality and Scalability

The central tension is between posterior fidelity and computational feasibility. Future work should report compute, memory, sampling cost, posterior diversity, and sensitivity to priors and approximations, not only downstream accuracy.

4.2 Evaluation Beyond Calibration Curves

Calibration curves and expected calibration error are useful but incomplete. Better evaluations should include proper scoring rules, OOD stress tests, subgroup reliability, selective prediction, abstention, active-learning efficiency, and decision utility.

4.3 Shift, Misspecification, and Data Quality

Bayesian notation does not guarantee robustness. If likelihoods, priors, architectures, or datasets are misspecified, uncertainty can remain misleading. Benchmarks should include realistic corruptions, temporal drift, covariate shift, label ambiguity, and annotation noise.

4.4 Human-Centered Uncertainty

Uncertainty must be communicated to users. Reports should explain what is uncertain, why it matters, how it affects the decision, and when escalation is required. Human-centered evaluation is especially

important in clinical, scientific, and safety-critical systems.

4.5 Toward Decision-Aware Bayesian Deep Learning

The field is moving from uncertainty as a diagnostic metric toward uncertainty as a decision resource. The most useful systems will connect approximate Bayesian inference, calibration, OOD detection, active data acquisition, and risk-aware deployment in one loop.

5 Conclusion

Bayesian deep learning provides a principled vocabulary for uncertainty quantification in neural systems, but practical success depends on approximations, evaluation, and deployment design. Strong methods must produce calibrated predictive distributions, expose epistemic uncertainty under shift, support decisions, and remain computationally viable. Progress will come from treating uncertainty as an end-to-end property of data, models, inference, metrics, and human action.

References

- [1] Yasser Abdullallah, Khalid A. Alobaid, Jason T. L. Wang, Haimin Wang, Vania K. Jordanova, Vasyl Yurchyshyn, Huseyin Cavus, and Ju Jing. Prediction of the SYM-H Index Using a Bayesian Deep Learning Method with Uncertainty Quantification. *arXiv preprint arXiv:2402.17196*, 2024. URL <https://arxiv.org/abs/2402.17196>.
- [2] K. Darshana Abeyrathna, Sara El Mekkaoui, Andreas Hafver, and Christian Agrell. The Probabilistic Tsetlin Machine: A Novel Approach to Uncertainty Quantification. *arXiv preprint arXiv:2410.17851*, 2024. URL <https://arxiv.org/abs/2410.17851>.
- [3] Kudaibergen Abutalip, Numan Saeed, Ikboljon Sobirov, Vincent Andrearczyk, Adrien Depursinge, and Mohammad Yaqub. EDUE: Expert Disagreement-Guided One-Pass Uncertainty Estimation for Medical Image Segmentation. *arXiv preprint arXiv:2403.16594*, 2024. URL <https://arxiv.org/abs/2403.16594>.
- [4] Jadie Adams, Riddhish Bhalodia, and Shireen Elhabian. Uncertain-DeepSSM: From Images to Probabilistic Shape Models. *arXiv preprint arXiv:2007.06516*, 2020. URL <https://arxiv.org/abs/2007.06516>.

- [5] Soyed Tuhin Ahmed. Scalable and Efficient Methods for Uncertainty Estimation and Reduction in Deep Learning. *arXiv preprint arXiv:2401.07145*, 2024. URL <https://arxiv.org/abs/2401.07145>.
- [6] Muhammad Ahtazaz Ahsan, Adnan Qayyum, Junaid Qadir, and Adeel Razi. An Active Learning Method for Diabetic Retinopathy Classification with Uncertainty Quantification. *arXiv preprint arXiv:2012.13325*, 2020. URL <https://arxiv.org/abs/2012.13325>.
- [7] Yusuke Akamatsu, Akinori F. Ebihara, and Terumi Umematsu. U-FaceBP: Uncertainty-aware Bayesian Ensemble Deep Learning for Face Video-based Blood Pressure Estimation. *arXiv preprint arXiv:2412.10679*, 2024. URL <https://arxiv.org/abs/2412.10679>.
- [8] Ahmed M. Alaa and Mihaela van der Schaar. Discriminative Jackknife: Quantifying Uncertainty in Deep Learning via Higher-Order Influence Functions. *arXiv preprint arXiv:2007.13481*, 2020. URL <https://arxiv.org/abs/2007.13481>.
- [9] Shahnawaz Alam, Mohammed Mudassir Uddin, and Mohammed Kaif Pasha. Constraint-Aware Neurosymbolic Uncertainty Quantification with Bayesian Deep Learning for Scientific Discovery. *arXiv preprint arXiv:2601.12442*, 2026. URL <https://arxiv.org/abs/2601.12442>.
- [10] Sarah I. Allec and Maxim Ziatdinov. Active and transfer learning with partially Bayesian neural networks for materials and chemicals. *arXiv preprint arXiv:2501.00952*, 2025. URL <https://arxiv.org/abs/2501.00952>.
- [11] Alexander Amini, Ava Soleimany, Sertac Karaman, and Daniela Rus. Spatial Uncertainty Sampling for End-to-End Control. *arXiv preprint arXiv:1805.04829*, 2018. URL <https://arxiv.org/abs/1805.04829>.
- [12] Hayk Amirkhanyan and Marco F. Huber. From Overfitting to Reliability: Introducing the Hierarchical Approximate Bayesian Neural Network. *arXiv preprint arXiv:2512.13111*, 2025. URL <https://arxiv.org/abs/2512.13111>.
- [13] Hamzeh Asgharnezhad, Afshar Shamsi, Roohallah Alizadehsani, Arash Mohammadi, and Hamid Alinejad-Rokny. Enhancing Monte Carlo Dropout Performance for Uncertainty Quantification. *arXiv preprint arXiv:2505.15671*, 2025. URL <https://arxiv.org/abs/2505.15671>.
- [14] Haimiti Atila and Seymour M. J. Spence. Deep Learning-Based Metamodeling of Nonlinear Stochastic Dynamic Systems under Parametric and Predictive Uncertainty. *arXiv preprint arXiv:2603.12012*, 2026. URL <https://arxiv.org/abs/2603.12012>.
- [15] Jincheng Bai, Qifan Song, and Guang Cheng. Efficient Variational Inference for Sparse Deep Learning with Theoretical Guarantee. *arXiv preprint arXiv:2011.07439*, 2020. URL <https://arxiv.org/abs/2011.07439>.
- [16] Luis Basora, Arthur Viens, Manuel Arias Chao, and Xavier Olive. A Benchmark on Uncertainty Quantification for Deep Learning Prognostics. *arXiv preprint arXiv:2302.04730*, 2023. URL <https://arxiv.org/abs/2302.04730>.
- [17] Simon Baur, Wojciech Samek, and Jackie Ma. Benchmarking Uncertainty and its Disentanglement in multi-label Chest X-Ray Classification. *arXiv preprint arXiv:2508.04457*, 2025. URL <https://arxiv.org/abs/2508.04457>.
- [18] Pavia Bera and Sanjukta Bhanja. Quantification of Uncertainties in Probabilistic Deep Neural Network by Implementing Boosting of Variational Inference. *arXiv preprint arXiv:2503.13909*, 2025. URL <https://arxiv.org/abs/2503.13909>.
- [19] Danyal F. Bhutto, Bo Zhu, Jeremiah Z. Liu, Neha Koonjoo, Hongwei B. Li, Bruce R. Rosen, and Matthew S. Rosen. Uncertainty Estimation and Out-of-Distribution Detection for Deep Learning-Based Image Reconstruction using the Local Lipschitz. *arXiv preprint arXiv:2305.07618*, 2023. URL <https://arxiv.org/abs/2305.07618>.
- [20] Shahin Boluki, Randy Ardywibowo, Siamak Zamani Dadaneh, Mingyuan Zhou, and Xiaoning Qian. Learnable Bernoulli Dropout for Bayesian Deep Learning. *arXiv preprint arXiv:2002.05155*, 2020. URL <https://arxiv.org/abs/2002.05155>.
- [21] Satya Borgohain, Klaus Ackermann, and Ruben Loaiza-Maya. Bayesian Neural Network Versus Ex-Post Calibration For Prediction Uncertainty. *arXiv preprint arXiv:2209.14594*, 2022. URL <https://arxiv.org/abs/2209.14594>.
- [22] Hendrik Borrás, Yong Wu, Bernhard Klein, and Holger Froning. Uncertainty-Preserving QBNNs: Multi-Level Quantization of SVI-Based Bayesian Neural Networks for Image Classification. *arXiv preprint arXiv:2512.10602*, 2025. URL <https://arxiv.org/abs/2512.10602>.
- [23] Luca Bortolussi, Francesca Cairolì, Ginevra Carbone, and Paolo Pulcini. Scalable Stochastic Parametric Verification with Stochastic Variational Smoothed Model Checking. *arXiv preprint arXiv:2205.05398*, 2022. URL <https://arxiv.org/abs/2205.05398>.
- [24] Nicolas Brosse, Carlos Riquelme, Alice Martin, Sylvain Gelly, and Eric Moulines. On Last-Layer Algorithms for Classification: Decoupling Representation from Uncertainty Estimation. *arXiv preprint arXiv:2001.08049*, 2020. URL <https://arxiv.org/abs/2001.08049>.

- [25] Kirill Bykov, Marina M. C. Hohne, Adelaide Creosteanu, Klaus-Robert Muller, Frederick Klauschen, Shinichi Nakajima, and Marius Kloft. Explaining Bayesian Neural Networks. *arXiv preprint arXiv:2108.10346*, 2021. URL <https://arxiv.org/abs/2108.10346>.
- [26] Ruizhe Cai, Ao Ren, Ning Liu, Caiwen Ding, Luhao Wang, Xuehai Qian, Massoud Pedram, and Yanzhi Wang. VIBNN: Hardware Acceleration of Bayesian Neural Networks. *arXiv preprint arXiv:1802.00822*, 2018. URL <https://arxiv.org/abs/1802.00822>.
- [27] Joao Caldeira and Brian Nord. Deeply Uncertain: Comparing Methods of Uncertainty Quantification in Deep Learning Algorithms. *arXiv preprint arXiv:2004.10710*, 2020. URL <https://arxiv.org/abs/2004.10710>.
- [28] Alejandro Calle-Saldarriaga, Felix Jimenez, Jack Grosskreuz, Jiazheng Wang, Jonathan Hobbs, and Matthias Katzfuss. Amortized Probabilistic Retrieval of Atmospheric CO₂ from OCO-2 Spectra Using Deep Learning with Laplace Approximations and Normalizing Flows. *arXiv preprint arXiv:2606.17413*, 2026. URL <https://arxiv.org/abs/2606.17413>.
- [29] Sergio Calvo-Ordóñez, Matthieu Meunier, Francesco Piatti, and Yuntao Shi. Partially Stochastic Infinitely Deep Bayesian Neural Networks. *arXiv preprint arXiv:2402.03495*, 2024. URL <https://arxiv.org/abs/2402.03495>.
- [30] Sergio Calvo-Ordóñez, Jonathan Plenck, Richard Bergna, Alvaro Cartea, Yarin Gal, Jose Miguel Hernandez-Lobato, and Kamil Ciosek. Richer Bayesian Last Layers with Subsampled NTK Features. *arXiv preprint arXiv:2602.01279*, 2026. URL <https://arxiv.org/abs/2602.01279>.
- [31] Michele Caprio, Souradeep Dutta, Kuk Jin Jang, Vivian Lin, Radoslav Ivanov, Oleg Sokolsky, and Insup Lee. Credal Bayesian Deep Learning. *arXiv preprint arXiv:2302.09656*, 2023. URL <https://arxiv.org/abs/2302.09656>.
- [32] Michele Caprio, Shireen K. Manchingal, and Fabio Cuzzolin. Credal and Interval Deep Evidential Classifications. *arXiv preprint arXiv:2512.05526*, 2025. URL <https://arxiv.org/abs/2512.05526>.
- [33] Luca Cardelli, Marta Kwiatkowska, Luca Laurenti, Nicola Paoletti, Andrea Patane, and Matthew Wicker. Statistical Guarantees for the Robustness of Bayesian Neural Networks. *arXiv preprint arXiv:1903.01980*, 2019. URL <https://arxiv.org/abs/1903.01980>.
- [34] Pietro Carloti, Nevena Gligic, and Arya Farahi. Density-Informed Pseudo-Counts for Calibrated Evidential Deep Learning. *arXiv preprint arXiv:2602.01477*, 2026. URL <https://arxiv.org/abs/2602.01477>.
- [35] Eduardo D C Carvalho, Ronald Clark, Andrea Nicastro, and Paul H J Kelly. Scalable Uncertainty for Computer Vision with Functional Variational Inference. *arXiv preprint arXiv:2003.03396*, 2020. URL <https://arxiv.org/abs/2003.03396>.
- [36] Seohyeon Cha, Honggu Kang, and Joonhyuk Kang. On the Temperature of Bayesian Graph Neural Networks for Conformal Prediction. *arXiv preprint arXiv:2310.11479*, 2023. URL <https://arxiv.org/abs/2310.11479>.
- [37] Rohitash Chandra and Joshua Simmons. Bayesian neural networks via MCMC: a Python-based tutorial. *arXiv preprint arXiv:2304.02595*, 2023. URL <https://arxiv.org/abs/2304.02595>.
- [38] Rohitash Chandra, Ayush Bhagat, Manavendra Maharana, and Pavel N. Krivitsky. Bayesian graph convolutional neural networks via tempered MCMC. *arXiv preprint arXiv:2104.08438*, 2021. URL <https://arxiv.org/abs/2104.08438>.
- [39] Mengyuan Chen, Junyu Gao, and Changsheng Xu. Revisiting Essential and Nonessential Settings of Evidential Deep Learning. *arXiv preprint arXiv:2410.00393*, 2024. URL <https://arxiv.org/abs/2410.00393>.
- [40] Yen-Chi Chen. A Frequentist Statistical Introduction to Variational Inference, Autoencoders, and Diffusion Models. *arXiv preprint arXiv:2510.18777*, 2025. URL <https://arxiv.org/abs/2510.18777>.
- [41] Yinsong Chen, Samson S. Yu, Zhong Li, and Chee Peng Lim. Addressing the Inconsistency in Bayesian Deep Learning via Generalized Laplace Approximation. *arXiv preprint arXiv:2405.13535*, 2024. URL <https://arxiv.org/abs/2405.13535>.
- [42] Ranjith Chodavarapu. Uncertainty-Aware Wildfire Smoke Density Classification from Satellite Imagery via CBAM-Augmented EfficientNet with Evidential Deep Learning. *arXiv preprint arXiv:2605.15894*, 2026. URL <https://arxiv.org/abs/2605.15894>.
- [43] Yongchan Chun, Chanhee Park, Jeongho Yoon, Jaehyung Seo, and Heuseok Lim. Evidential Transformation Network: Turning Pretrained Models into Evidential Models for Post-hoc Uncertainty Estimation. *arXiv preprint arXiv:2604.08627*, 2026. URL <https://arxiv.org/abs/2604.08627>.
- [44] Tristan Cinquin and Robert Bamler. Regularized KL-Divergence for Well-Defined Function-Space Variational Inference in Bayesian neural networks. *arXiv preprint arXiv:2406.04317*, 2024. URL <https://arxiv.org/abs/2406.04317>.

- [45] Tristan Cinquin, Alexander Immer, Max Horn, and Vincent Fortuin. Pathologies in priors and inference for Bayesian transformers. *arXiv preprint arXiv:2110.04020*, 2021. URL <https://arxiv.org/abs/2110.04020>.
- [46] Tristan Cinquin, Marvin Pfortner, Vincent Fortuin, Philipp Hennig, and Robert Bamler. FSP-Laplace: Function-Space Priors for the Laplace Approximation in Bayesian Deep Learning. *arXiv preprint arXiv:2407.13711*, 2024. URL <https://arxiv.org/abs/2407.13711>.
- [47] Nastaran Dabiran, Brandon Robinson, Rimple Sandhu, Mohammad Khalil, Dominique Poirel, and Abhijit Sarkar. Sparse Bayesian neural networks for regression: Tackling overfitting and computational challenges in uncertainty quantification. *arXiv preprint arXiv:2310.15614*, 2023. URL <https://arxiv.org/abs/2310.15614>.
- [48] Joel Janek Dabrowski and Daniel Edward Pagnam. Bayesian Neural Network Inference via Implicit Models and the Posterior Predictive Distribution. *arXiv preprint arXiv:2209.02188*, 2022. URL <https://arxiv.org/abs/2209.02188>.
- [49] A. Daneshkhah, O. Chatrabgoun, M. Esmaeilbeigi, T. Sedighi, and S. Abolfathi. Uncertainty Quantification of Darcy Flow through Porous Media using Deep Gaussian Process. *arXiv preprint arXiv:2011.01647*, 2020. URL <https://arxiv.org/abs/2011.01647>.
- [50] Rodrigo Pereira David. Tubular Riemannian Laplace Approximations for Bayesian Neural Networks. *arXiv preprint arXiv:2512.24381*, 2025. URL <https://arxiv.org/abs/2512.24381>.
- [51] Cai Davies, Marc Roig Vilamala, Alun D. Preece, Federico Cerutti, Lance M. Kaplan, and Supriyo Chakraborty. Knowledge from Uncertainty in Evidential Deep Learning. *arXiv preprint arXiv:2310.12663*, 2023. URL <https://arxiv.org/abs/2310.12663>.
- [52] Erik Daxberger, Eric Nalisnick, James Uquhart Allingham, Javier Antoran, and Jose Miguel Hernandez-Lobato. Bayesian Deep Learning via Subnetwork Inference. *arXiv preprint arXiv:2010.14689*, 2020. URL <https://arxiv.org/abs/2010.14689>.
- [53] Erik Daxberger, Agustinus Kristiadi, Alexander Immer, Runa Eschenhagen, Matthias Bauer, and Philipp Hennig. Laplace Redux – Effortless Bayesian Deep Learning. *arXiv preprint arXiv:2106.14806*, 2021. URL <https://arxiv.org/abs/2106.14806>.
- [54] Feras Dayoub, Niko Sunderhauf, and Peter Corke. Episode-Based Active Learning with Bayesian Neural Networks. *arXiv preprint arXiv:1703.07473*, 2017. URL <https://arxiv.org/abs/1703.07473>.
- [55] Sajib Debnath and Md. Uzzal Mia. Bayesian Transformer for Probabilistic Load Forecasting in Smart Grids. *arXiv preprint arXiv:2603.07899*, 2026. URL <https://arxiv.org/abs/2603.07899>.
- [56] Kumari Deepshikha, Sai Harsha Yelleni, P. K. Sri-jith, and C Krishna Mohan. Monte Carlo Drop-Block for Modelling Uncertainty in Object Detection. *arXiv preprint arXiv:2108.03614*, 2021. URL <https://arxiv.org/abs/2108.03614>.
- [57] Zhijie Deng, Yucen Luo, Jun Zhu, and Bo Zhang. Measuring Uncertainty through Bayesian Learning of Deep Neural Network Structure. *arXiv preprint arXiv:1911.09804*, 2019. URL <https://arxiv.org/abs/1911.09804>.
- [58] Ratneel Deo, Scott Sisson, Jody M. Webster, and Rohitash Chandra. Compact Bayesian Neural Networks via pruned MCMC sampling. *arXiv preprint arXiv:2501.06962*, 2025. URL <https://arxiv.org/abs/2501.06962>.
- [59] Stefan Depeweg, Jose Miguel Hernandez-Lobato, Finale Doshi-Velez, and Steffen Udluft. Uncertainty Decomposition in Bayesian Neural Networks with Latent Variables. *arXiv preprint arXiv:1706.08495*, 2017. URL <https://arxiv.org/abs/1706.08495>.
- [60] Arnaud Descours, Tom Huix, Arnaud Guillin, Manon Michel, Eric Moulines, and Boris Nectoux. Law of Large Numbers for Bayesian two-layer Neural Network trained with Variational Inference. *arXiv preprint arXiv:2307.04779*, 2023. URL <https://arxiv.org/abs/2307.04779>.
- [61] Gianluca Detommaso, Alberto Gasparin, Andrew Wilson, and Cedric Archambeau. Uncertainty Calibration in Bayesian Neural Networks via Distance-Aware Priors. *arXiv preprint arXiv:2207.08200*, 2022. URL <https://arxiv.org/abs/2207.08200>.
- [62] Simon Dirmeier and Fernando Perez-Cruz. Diffusion models for probabilistic programming. *arXiv preprint arXiv:2311.00474*, 2023. URL <https://arxiv.org/abs/2311.00474>.
- [63] Aslak Djupskas, Alexander Johannes Stasik, and Signe Riemer-Srensen. Unreliable Uncertainty Estimates with Monte Carlo Dropout. *arXiv preprint arXiv:2512.14851*, 2025. URL <https://arxiv.org/abs/2512.14851>.
- [64] Ruxiao Duan, Brian Caffo, Harrison X. Bai, Haris I. Sair, and Craig Jones. Evidential Uncertainty Quantification: A Variance-Based Perspective. *arXiv preprint arXiv:2311.11367*, 2023. URL <https://arxiv.org/abs/2311.11367>.
- [65] Canberk Ekmekci and Mujdat Cetin. Uncertainty Quantification for Deep Unrolling-Based Computational Imaging. *arXiv preprint arXiv:2207.00698*, 2022. URL <https://arxiv.org/abs/2207.00698>.

- [66] Clement Etienam, Siying Shen, Edward J O'Dwyer, and Joshua Sykes. A Novel Cluster Classify Regress Model Predictive Controller Formulation; CCR-MPC. *arXiv preprint arXiv:2101.07655*, 2021. URL <https://arxiv.org/abs/2101.07655>.
- [67] Josua Faller and Jorg Martin. Low Rank Based Subspace Inference for the Laplace Approximation of Bayesian Neural Networks. *arXiv preprint arXiv:2502.02345*, 2025. URL <https://arxiv.org/abs/2502.02345>.
- [68] Da Fan, David John Gagne, Steven J. Greybush, Eugene E. Clothiaux, John S. Schreck, and Chaopeng Shen. Bayesian Deep Learning for Convective Initiation Nowcasting Uncertainty Estimation. *arXiv preprint arXiv:2507.16219*, 2025. URL <https://arxiv.org/abs/2507.16219>.
- [69] Huajian Fang, Dennis Becker, Stefan Wermter, and Timo Gerkmann. Integrating Uncertainty into Neural Network-based Speech Enhancement. *arXiv preprint arXiv:2305.08744*, 2023. URL <https://arxiv.org/abs/2305.08744>.
- [70] Riccardo Farris, Emanuele Telari, Nongnuch Artrith, Konstantin Neyman, and Albert Bruix. Bayesian Neural Networks versus deep ensembles for uncertainty quantification in machine learning interatomic potentials. *arXiv preprint arXiv:2509.19180*, 2025. URL <https://arxiv.org/abs/2509.19180>.
- [71] Yassir Fathullah, Guoxuan Xia, and Mark Gales. Logit-Based Ensemble Distribution Distillation for Robust Autoregressive Sequence Uncertainties. *arXiv preprint arXiv:2305.10384*, 2023. URL <https://arxiv.org/abs/2305.10384>.
- [72] Abdur R. Fayjie, Jutika Borah, Florencia Carbone, Jan Tack, and Patrick Vandewalle. Predictive uncertainty estimation in deep learning for lung carcinoma classification in digital pathology under real dataset shifts. *arXiv preprint arXiv:2408.08432*, 2024. URL <https://arxiv.org/abs/2408.08432>.
- [73] Mohammed Fellaji, Frederic Pennerath, Briec Conan-Guez, and Miguel Couceiro. On the Calibration of Epistemic Uncertainty: Principles, Paradoxes and Conflictual Loss. *arXiv preprint arXiv:2407.12211*, 2024. URL <https://arxiv.org/abs/2407.12211>.
- [74] Brandon R. Feng, David Keetae Park, Xihaier Luo, Arantxa Urdangarin, Shinjae Yoo, and Brian J. Reich. STACI: Spatio-Temporal Aleatoric Conformal Inference. *arXiv preprint arXiv:2505.21658*, 2025. URL <https://arxiv.org/abs/2505.21658>.
- [75] Jianxiang Feng, Maximilian Durner, Zoltan-Csaba Marton, Ferenc Balint-Benczedi, and Rudolph Triebel. Introspective Robot Perception using Smoothed Predictions from Bayesian Neural Networks. *arXiv preprint arXiv:2109.12869*, 2021. URL <https://arxiv.org/abs/2109.12869>.
- [76] John Fischer, Marko Orescanin, and Eric Eckstrand. VI-PANN: Harnessing Transfer Learning and Uncertainty-Aware Variational Inference for Improved Generalization in Audio Pattern Recognition. *arXiv preprint arXiv:2401.05531*, 2024. URL <https://arxiv.org/abs/2401.05531>.
- [77] John Fischer, Marko Orescanin, Justin Loomis, and Patrick McClure. Federated Bayesian Deep Learning: The Application of Statistical Aggregation Methods to Bayesian Models. *arXiv preprint arXiv:2403.15263*, 2024. URL <https://arxiv.org/abs/2403.15263>.
- [78] Andrew Y. K. Foong, David R. Burt, Yingzhen Li, and Richard E. Turner. On the Expressiveness of Approximate Inference in Bayesian Neural Networks. *arXiv preprint arXiv:1909.00719*, 2019. URL <https://arxiv.org/abs/1909.00719>.
- [79] Andrew Y. K. Foong, Yingzhen Li, Jose Miguel Hernandez-Lobato, and Richard E. Turner. 'In-Between' Uncertainty in Bayesian Neural Networks. *arXiv preprint arXiv:1906.11537*, 2019. URL <https://arxiv.org/abs/1906.11537>.
- [80] Courtney Franzen and Farhad Pourkamali-Anaraki. Ensemble-Based Dirichlet Modeling for Predictive Uncertainty and Selective Classification. *arXiv preprint arXiv:2604.06032*, 2026. URL <https://arxiv.org/abs/2604.06032>.
- [81] Wolfgang Fruehwirt, Adam D. Cobb, Martin Mairhofer, Leonard Weydemann, Heinrich Garn, Reinhold Schmidt, Thomas Benke, Peter Dal-Bianco, Gerhard Ransmayr, Markus Waser, Dieter Grossegger, Pengfei Zhang, Georg Dorffner, and Stephen Roberts. Bayesian deep neural networks for low-cost neurophysiological markers of Alzheimer's disease severity. *arXiv preprint arXiv:1812.04994*, 2018. URL <https://arxiv.org/abs/1812.04994>.
- [82] Futoshi Futami, Tomoharu Iwata, Naonori Ueda, Issei Sato, and Masashi Sugiyama. Excess risk analysis for epistemic uncertainty with application to variational inference. *arXiv preprint arXiv:2206.01606*, 2022. URL <https://arxiv.org/abs/2206.01606>.
- [83] Yarin Gal, Riashat Islam, and Zoubin Ghahramani. Deep Bayesian Active Learning with Image Data. *arXiv preprint arXiv:1703.02910*, 2017. URL <https://arxiv.org/abs/1703.02910>.
- [84] Tewodros Syum Gebre, Jagrati Talreja, Matilda Anokye, and Leila Hashemi-Beni. Overcoming "Physics Shock" in Earth Observation A Heteroscedastic Uncertainty Framework for PINN-based Flood Inference. *arXiv*

- preprint *arXiv:2605.24106*, 2026. URL <https://arxiv.org/abs/2605.24106>.
- [85] Nicholas Geneva and Nicholas Zabaras. Quantifying model form uncertainty in Reynolds-averaged turbulence models with Bayesian deep neural networks. *arXiv preprint arXiv:1807.02901*, 2018. URL <https://arxiv.org/abs/1807.02901>.
- [86] Javad Ghorbanian, Nicholas Casaprima, and Audrey Olivier. Empowering Bayesian Neural Networks with Functional Priors through Anchored Ensembling for Mechanics Surrogate Modeling Applications. *arXiv preprint arXiv:2409.05234*, 2024. URL <https://arxiv.org/abs/2409.05234>.
- [87] Riddhi Pratim Ghosh and Ian Barnett. Bayesian Neural Networks vs. Mixture Density Networks: Theoretical and Empirical Insights for Uncertainty-Aware Nonlinear Modeling. *arXiv preprint arXiv:2510.25001*, 2025. URL <https://arxiv.org/abs/2510.25001>.
- [88] Misha Glazunov and Apostolis Zarras. Do Bayesian Variational Autoencoders Know What They Don't Know? *arXiv preprint arXiv:2212.14272*, 2022. URL <https://arxiv.org/abs/2212.14272>.
- [89] Ethan Goan, Dimitri Perrin, Kerrie Mengersen, and Clinton Fookes. Piecewise Deterministic Markov Processes for Bayesian Neural Networks. *arXiv preprint arXiv:2302.08724*, 2023. URL <https://arxiv.org/abs/2302.08724>.
- [90] Aurora Grefsrud, Nello Blaser, and Trygve Buanes. Calibrated and uncertain? Evaluating uncertainty estimates in binary classification models. *arXiv preprint arXiv:2508.11460*, 2025. URL <https://arxiv.org/abs/2508.11460>.
- [91] Ruben Grewal, Paolo Tonella, and Andrea Stocco. Predicting Safety Misbehaviours in Autonomous Driving Systems using Uncertainty Quantification. *arXiv preprint arXiv:2404.18573*, 2024. URL <https://arxiv.org/abs/2404.18573>.
- [92] Bjarne Grimstad, Mathilde Hotvedt, Anders T. Sandnes, Odd Kolbjørnsen, and Lars S. Imsland. Bayesian Neural Networks for Virtual Flow Metering: An Empirical Study. *arXiv preprint arXiv:2102.01391*, 2021. URL <https://arxiv.org/abs/2102.01391>.
- [93] Daniel Habermann, Marvin Schmitt, Lars Kuhmichel, Andreas Bulling, Stefan T. Radev, and Paul-Christian Burkner. Amortized Bayesian Multilevel Models. *arXiv preprint arXiv:2408.13230*, 2024. URL <https://arxiv.org/abs/2408.13230>.
- [94] Farida Al Haddad, Yuxin Wang, and Malcolm Miele. UAC: Uncertainty-Aware Calibration of Neural Networks for Gesture Detection. *arXiv preprint arXiv:2504.02895*, 2025. URL <https://arxiv.org/abs/2504.02895>.
- [95] Pengcheng Hao, Huaize Tang, Ercan Engin Kuruoglu, and Wenbo Ding. Function-Space Empirical Bayes Regularisation with Large Vision-Language Model Priors. *arXiv preprint arXiv:2602.03119*, 2026. URL <https://arxiv.org/abs/2602.03119>.
- [96] Arman Hasanzadeh, Ehsan Hajiramezanali, Shahin Boluki, Mingyuan Zhou, Nick Duffield, Krishna Narayanan, and Xiaoning Qian. Bayesian Graph Neural Networks with Adaptive Connection Sampling. *arXiv preprint arXiv:2006.04064*, 2020. URL <https://arxiv.org/abs/2006.04064>.
- [97] Manuel Haussmann, Fred A. Hamprecht, and Melih Kandemir. Sampling-Free Variational Inference of Bayesian Neural Networks by Variance Backpropagation. *arXiv preprint arXiv:1805.07654*, 2018. URL <https://arxiv.org/abs/1805.07654>.
- [98] Manuel Haussmann, Sebastian Gerwinn, and Melih Kandemir. Bayesian Evidential Deep Learning with PAC Regularization. *arXiv preprint arXiv:1906.00816*, 2019. URL <https://arxiv.org/abs/1906.00816>.
- [99] Berk Hayta, Hannah Laus, Simon Mittermaier, and Felix Krahmer. Plug-in Losses for Evidential Deep Learning: A Simplified Framework for Uncertainty Estimation that Includes the Softmax Classifier. *arXiv preprint arXiv:2605.22746*, 2026. URL <https://arxiv.org/abs/2605.22746>.
- [100] Conor Heins, Hao Wu, Dimitrije Markovic, Alexander Tschantz, Jeff Beck, and Christopher Buckley. Gradient-free variational learning with conditional mixture networks. *arXiv preprint arXiv:2408.16429*, 2024. URL <https://arxiv.org/abs/2408.16429>.
- [101] Rouaa Hoblos, Noura Dridi, Nouredine Zerhouni, and Zeina Al Masry. Dirichlet-Based Monte Carlo Dropout for Uncertainty Estimation in Neural Networks. *arXiv preprint arXiv:2605.23635*, 2026. URL <https://arxiv.org/abs/2605.23635>.
- [102] Xiaoling Hu, Karthik Gopinath, Peirong Liu, Malte Hoffmann, Koen Van Leemput, Oula Puonti, and Juan Eugenio Iglesias. Hierarchical Uncertainty Estimation for Learning-based Registration in Neuroimaging. *arXiv preprint arXiv:2410.09299*, 2024. URL <https://arxiv.org/abs/2410.09299>.
- [103] Po-Yu Huang, Wan-Ting Hsu, Chun-Yueh Chiu, Ting-Fan Wu, and Min Sun. Efficient Uncertainty Estimation for Semantic Segmentation in Videos. *arXiv preprint arXiv:1807.11037*, 2018. URL <https://arxiv.org/abs/1807.11037>.
- [104] Matthias Humt, Jongseok Lee, and Rudolph

- Triebel. Bayesian Optimization Meets Laplace Approximation for Robotic Introspection. *arXiv preprint arXiv:2010.16141*, 2020. URL <https://arxiv.org/abs/2010.16141>.
- [105] Denis Huseljic, Marek Herde, Lukas Rauch, Paul Hahn, Zhixin Huang, Daniel Kottke, Stephan Vogt, and Bernhard Sick. Efficient Bayesian Updates for Deep Active Learning via Laplace Approximations. *arXiv preprint arXiv:2210.06112*, 2022. URL <https://arxiv.org/abs/2210.06112>.
- [106] Eirik Hyheim, Lars Skaaret-Lund, Solve Sb, and Aliaksandr Hubin. Explainable Bayesian deep learning through input-skip Latent Binary Bayesian Neural Networks. *arXiv preprint arXiv:2503.10496*, 2025. URL <https://arxiv.org/abs/2503.10496>.
- [107] Alexander Immer. Disentangling the Gauss-Newton Method and Approximate Inference for Neural Networks. *arXiv preprint arXiv:2007.11994*, 2020. URL <https://arxiv.org/abs/2007.11994>.
- [108] Alexander Immer, Maciej Korzepa, and Matthias Bauer. Improving predictions of Bayesian neural nets via local linearization. *arXiv preprint arXiv:2008.08400*, 2020. URL <https://arxiv.org/abs/2008.08400>.
- [109] Keito Inoshita and Takato Ueno. Uncertainty Decomposition via Cyclical SG-MCMC and Soft-label Learning for Subjective NLP. *arXiv preprint arXiv:2605.24773*, 2026. URL <https://arxiv.org/abs/2605.24773>.
- [110] Masha Itkina and Mykel J. Kochenderfer. Interpretable Self-Aware Neural Networks for Robust Trajectory Prediction. *arXiv preprint arXiv:2211.08701*, 2022. URL <https://arxiv.org/abs/2211.08701>.
- [111] Pavel Izmailov, Sharad Vikram, Matthew D. Hoffman, and Andrew Gordon Wilson. What Are Bayesian Neural Network Posteriors Really Like? *arXiv preprint arXiv:2104.14421*, 2021. URL <https://arxiv.org/abs/2104.14421>.
- [112] Reza Jalayer, Masoud Jalayer, Andrea Mor, Carlotta Orsenigo, and Carlo Vercellis. Evaluating deep learning models for fault diagnosis of a rotating machinery with epistemic and aleatoric uncertainty. *arXiv preprint arXiv:2412.18980*, 2024. URL <https://arxiv.org/abs/2412.18980>.
- [113] Benben Jiang, Yixing Wang, Zhenghua Ma, and Qiugang Lu. Fast Charging of Lithium-Ion Batteries Using Deep Bayesian Optimization with Recurrent Neural Network. *arXiv preprint arXiv:2304.04195*, 2023. URL <https://arxiv.org/abs/2304.04195>.
- [114] Pedro Jimenez, Luis A. Ortega, Pablo Morales-Alvarez, and Daniel Hernandez-Lobato. Improving the Linearized Laplace Approximation via Quadratic Approximations. *arXiv preprint arXiv:2602.03394*, 2026. URL <https://arxiv.org/abs/2602.03394>.
- [115] Peter Van Katwyk and Karianne J. Bergen. HybridFlow: Quantification of Aleatoric and Epistemic Uncertainty with a Single Hybrid Model. *arXiv preprint arXiv:2510.05054*, 2025. URL <https://arxiv.org/abs/2510.05054>.
- [116] Devinder Kaur, Shama Naz Islam, Md. Apel Mahmud, Md. Enamul Haque, and Adnan Anwar. A VAE-Bayesian Deep Learning Scheme for Solar Generation Forecasting based on Dimensionality Reduction. *arXiv preprint arXiv:2103.12969*, 2021. URL <https://arxiv.org/abs/2103.12969>.
- [117] Eiji Kawasaki, Markus Holzmann, and Lawrence Adu-Gyamfi. Data Subsampling for Bayesian Neural Networks. *arXiv preprint arXiv:2210.09141*, 2022. URL <https://arxiv.org/abs/2210.09141>.
- [118] Xiongwen Ke and Yanan Fan. On the optimization and pruning for Bayesian deep learning. *arXiv preprint arXiv:2210.12957*, 2022. URL <https://arxiv.org/abs/2210.12957>.
- [119] Vahid Keshavarzzadeh, Robert M. Kirby, and Akil Narayan. Variational Inference for Nonlinear Inverse Problems via Neural Net Kernels: Comparison to Bayesian Neural Networks, Application to Topology Optimization. *arXiv preprint arXiv:2205.03681*, 2022. URL <https://arxiv.org/abs/2205.03681>.
- [120] Mohammad Emtiyaz Khan, Didrik Nielsen, Voot Tangkaratt, Wu Lin, Yarin Gal, and Akash Srivastava. Fast and Scalable Bayesian Deep Learning by Weight-Perturbation in Adam. *arXiv preprint arXiv:1806.04854*, 2018. URL <https://arxiv.org/abs/1806.04854>.
- [121] Maryam Kheirandish, Shengfan Zhang, Donald G. Catanzaro, and Valeriu Crudu. Quantifying Uncertainty in Deep Learning Classification with Noise in Discrete Inputs for Risk-Based Decision Making. *arXiv preprint arXiv:2310.06105*, 2023. URL <https://arxiv.org/abs/2310.06105>.
- [122] Ayush Khot, Xiwei Wang, Avik Roy, Volodymyr Kindratenko, and Mark S. Neubauer. Evidential Deep Learning for Uncertainty Quantification and Out-of-Distribution Detection in Jet Identification using Deep Neural Networks. *arXiv preprint arXiv:2501.05656*, 2025. URL <https://arxiv.org/abs/2501.05656>.
- [123] Minyoung Kim. Large-scale Score-based Variational Posterior Inference for Bayesian Deep Neural Networks. *arXiv preprint arXiv:2602.05873*, 2026. URL <https://arxiv.org/abs/2602.05873>.
- [124] Namjung Kim, Dongseok Lee, Jongbin Yu,

- Sung Woong Cho, Dosung Lee, Yesol Park, and Youngjoon Hong. Toward a Robust and Generalizable Metamaterial Foundation Model. *arXiv preprint arXiv:2507.02436*, 2025. URL <https://arxiv.org/abs/2507.02436>.
- [125] QHwan Kim, Joon-Hyuk Ko, Sunghoon Kim, Nojun Park, and Wonho Jhe. Bayesian neural network with pretrained protein embedding enhances prediction accuracy of drug-protein interaction. *arXiv preprint arXiv:2012.08194*, 2020. URL <https://arxiv.org/abs/2012.08194>.
- [126] Sehwan Kim, Qifan Song, and Faming Liang. Stochastic Gradient Langevin Dynamics Algorithms with Adaptive Drifts. *arXiv preprint arXiv:2009.09535*, 2020. URL <https://arxiv.org/abs/2009.09535>.
- [127] Charlie Kirkwood, Theo Economou, and Nicolas Pugeault. Bayesian deep learning for mapping via auxiliary information: a new era for geostatistics? *arXiv preprint arXiv:2008.07320*, 2020. URL <https://arxiv.org/abs/2008.07320>.
- [128] Andreas Kla, Sven M. Lorenz, Martin W. Lauer-Schmaltz, David Rugamer, Bernd Bischl, Christopher Mutschler, and Felix Ott. Uncertainty-aware Evaluation of Time-Series Classification for On-line Handwriting Recognition with Domain Shift. *arXiv preprint arXiv:2206.08640*, 2022. URL <https://arxiv.org/abs/2206.08640>.
- [129] Bernhard Klein. Resource-Efficient and Robust Inference of Deep and Bayesian Neural Networks on Embedded and Analog Computing Platforms. *arXiv preprint arXiv:2510.24951*, 2025. URL <https://arxiv.org/abs/2510.24951>.
- [130] Bernhard Klein, Falk Selker, Hendrik Borrás, Sophie Steger, Franz Pernkopf, and Holger Froning. Accelerated Execution of Bayesian Neural Networks using a Single Probabilistic Forward Pass and Code Generation. *arXiv preprint arXiv:2511.23440*, 2025. URL <https://arxiv.org/abs/2511.23440>.
- [131] Jeremias Knoblauch, Jack Jewson, and Theodoros Damoulas. Generalized Variational Inference: Three arguments for deriving new Posteriors. *arXiv preprint arXiv:1904.02063*, 2019. URL <https://arxiv.org/abs/1904.02063>.
- [132] Brandon Kriesten and T. J. Hobbs. Anomalous electroweak physics unraveled via evidential deep learning. *arXiv preprint arXiv:2412.16286*, 2024. URL <https://arxiv.org/abs/2412.16286>.
- [133] Ranganath Krishnan, Mahesh Subedar, and Omesh Tickoo. BAR: Bayesian Activity Recognition using variational inference. *arXiv preprint arXiv:1811.03305*, 2018. URL <https://arxiv.org/abs/1811.03305>.
- [134] Agustinus Kristiadi, Sina Daubener, and Asja Fischer. Predictive Uncertainty Quantification with Compound Density Networks. *arXiv preprint arXiv:1902.01080*, 2019. URL <https://arxiv.org/abs/1902.01080>.
- [135] Agustinus Kristiadi, Alexander Immer, Runa Eschenhagen, and Vincent Fortuin. Promises and Pitfalls of the Linearized Laplace in Bayesian Optimization. *arXiv preprint arXiv:2304.08309*, 2023. URL <https://arxiv.org/abs/2304.08309>.
- [136] Fabian Kupperts, Jan Kronenberger, Jonas Schneider, and Anselm Haselhoff. Bayesian Confidence Calibration for Epistemic Uncertainty Modelling. *arXiv preprint arXiv:2109.10092*, 2021. URL <https://arxiv.org/abs/2109.10092>.
- [137] Wai Meng Kwok, Sarat Chandra Dass, and George Streftaris. Deep Learning Aided Laplace Based Bayesian Inference for Epidemiological Systems. *arXiv preprint arXiv:2210.08865*, 2022. URL <https://arxiv.org/abs/2210.08865>.
- [138] Tyler LaBonte, Carianne Martinez, and Scott A. Roberts. We Know Where We Don't Know: 3D Bayesian CNNs for Credible Geometric Uncertainty. *arXiv preprint arXiv:1910.10793*, 2019. URL <https://arxiv.org/abs/1910.10793>.
- [139] George Lamb and Brooks Paige. Bayesian Graph Neural Networks for Molecular Property Prediction. *arXiv preprint arXiv:2012.02089*, 2020. URL <https://arxiv.org/abs/2012.02089>.
- [140] Olivier Laurent, Emanuel Aldea, and Gianni Franchi. A Symmetry-Aware Exploration of Bayesian Neural Network Posteriors. *arXiv preprint arXiv:2310.08287*, 2023. URL <https://arxiv.org/abs/2310.08287>.
- [141] Taekbeom Lee, Dabin Kim, Youngseok Jang, and H. Jin Kim. HERE: Hierarchical Active Exploration of Radiance Field with Epistemic Uncertainty Minimization. *arXiv preprint arXiv:2601.07242*, 2026. URL <https://arxiv.org/abs/2601.07242>.
- [142] Shiye Lei, Zhuozhuo Tu, Leszek Rutkowski, Feng Zhou, Li Shen, Fengxiang He, and Dacheng Tao. Spatial-Temporal-Fusion BNN: Variational Bayesian Feature Layer. *arXiv preprint arXiv:2112.06281*, 2021. URL <https://arxiv.org/abs/2112.06281>.
- [143] Junbo Li, Zichen Miao, Qiang Qiu, and Ruqi Zhang. Training Bayesian Neural Networks with Sparse Subspace Variational Inference. *arXiv preprint arXiv:2402.11025*, 2024. URL <https://arxiv.org/abs/2402.11025>.
- [144] Longze Li, Jiang Chang, Aleksandar Vakanski, Yachun Wang, Tiankai Yao, and Min Xian. Uncertainty Quantification in Multivariable Regression

- for Material Property Prediction with Bayesian Neural Networks. *arXiv preprint arXiv:2311.02495*, 2023. URL <https://arxiv.org/abs/2311.02495>.
- [145] Yucen Lily Li, Tim G. J. Rudner, and Andrew Gordon Wilson. A Study of Bayesian Neural Network Surrogates for Bayesian Optimization. *arXiv preprint arXiv:2305.20028*, 2023. URL <https://arxiv.org/abs/2305.20028>.
- [146] Haojie Lian, Xinhao Li, Yilin Qu, Jing Du, Zhuxuan Meng, Jie Liu, and Leilei Chen. Bayesian uncertainty analysis for underwater 3D reconstruction with neural radiance fields. *arXiv preprint arXiv:2407.08154*, 2024. URL <https://arxiv.org/abs/2407.08154>.
- [147] Xinzhu Liang, Joseph M. Lukens, Sanjaya Lohani, Brian T. Kirby, Thomas A. Searles, Xin Qiu, and Kody J. H. Law. Scalable Bayesian Monte Carlo: fast uncertainty estimation beyond deep ensembles. *arXiv preprint arXiv:2505.13585*, 2025. URL <https://arxiv.org/abs/2505.13585>.
- [148] Luca Della Libera, Jacopo Andreoli, Davide Dalle Pezze, Mirco Ravanelli, and Gian Antonio Susto. Bayesian Deep Learning for Remaining Useful Life Estimation via Stein Variational Gradient Descent. *arXiv preprint arXiv:2402.01098*, 2024. URL <https://arxiv.org/abs/2402.01098>.
- [149] Moule Lin, Shuhao Guan, Weipeng Jing, Goetz Botterweck, and Andrea Patane. Stochastic Weight Sharing for Bayesian Neural Networks. *arXiv preprint arXiv:2505.17856*, 2025. URL <https://arxiv.org/abs/2505.17856>.
- [150] Adrian Lindenmeyer, Malte Blattmann, Stefan Franke, Thomas Neumuth, and Daniel Schneider. Inadequacy of common stochastic neural networks for reliable clinical decision support. *arXiv preprint arXiv:2401.13657*, 2024. URL <https://arxiv.org/abs/2401.13657>.
- [151] Jeremiah Zhe Liu, Zi Lin, Shreyas Padhy, Dustin Tran, Tania Bedrax-Weiss, and Balaji Lakshminarayanan. Simple and Principled Uncertainty Estimation with Deterministic Deep Learning via Distance Awareness. *arXiv preprint arXiv:2006.10108*, 2020. URL <https://arxiv.org/abs/2006.10108>.
- [152] Shengjie Kris Liu, Siqin Wang, and Lu Zhang. Uncertainty-Aware Hourly Air Temperature Mapping at 2 km Resolution via Physics-Guided Deep Learning. *arXiv preprint arXiv:2509.12329*, 2025. URL <https://arxiv.org/abs/2509.12329>.
- [153] Yehao Liu, Matteo Pagliardini, Tatjana Chavdarova, and Sebastian U. Stich. The Peril of Popular Deep Learning Uncertainty Estimation Methods. *arXiv preprint arXiv:2112.05000*, 2021. URL <https://arxiv.org/abs/2112.05000>.
- [154] Yijia Liu and Xiao Wang. Implicit Generative Prior for Bayesian Neural Networks. *arXiv preprint arXiv:2404.18008*, 2024. URL <https://arxiv.org/abs/2404.18008>.
- [155] Miguel Lopez-Perez, Marco Miani, Valery Naranjo, Sren Hauberg, and Aasa Feragen. Bayesian generative models can flag performance loss, bias, and out-of-distribution image content. *arXiv preprint arXiv:2503.17477*, 2025. URL <https://arxiv.org/abs/2503.17477>.
- [156] Lars Lorch, Jonas Rothfuss, Bernhard Scholkopf, and Andreas Krause. DiBS: Differentiable Bayesian Structure Learning. *arXiv preprint arXiv:2105.11839*, 2021. URL <https://arxiv.org/abs/2105.11839>.
- [157] Samuel Yanes Luis, Alejandro Casado Perez, Alejandro Mendoza Barrionuevo, Dame Seck Diop, Sergio Toral Marin, and Daniel Gutierrez Reina. Uncertainty Estimation for Deep Reconstruction in Actuatric Disaster Scenarios with Autonomous Vehicles. *arXiv preprint arXiv:2604.06387*, 2026. URL <https://arxiv.org/abs/2604.06387>.
- [158] Xihaier Luo and Ahsan Kareem. Bayesian deep learning with hierarchical prior: Predictions from limited and noisy data. *arXiv preprint arXiv:1907.04240*, 2019. URL <https://arxiv.org/abs/1907.04240>.
- [159] Yadan Luo, Zi Huang, Zheng Zhang, Ziwei Wang, Mahsa Baktashmotlagh, and Yang Yang. Learning from the Past: Continual Meta-Learning via Bayesian Graph Modeling. *arXiv preprint arXiv:1911.04695*, 2019. URL <https://arxiv.org/abs/1911.04695>.
- [160] Qing Lyu, Christopher T. Whitlow, and Ge Wang. SoftDropConnect (SDC) – Effective and Efficient Quantification of the Network Uncertainty in Deep MR Image Analysis. *arXiv preprint arXiv:2201.08418*, 2022. URL <https://arxiv.org/abs/2201.08418>.
- [161] Rufe Ma and Chao Chen. RF-BayesPhysNet: A Bayesian rPPG Uncertainty Estimation Method for Complex Scenarios. *arXiv preprint arXiv:2504.03915*, 2025. URL <https://arxiv.org/abs/2504.03915>.
- [162] Wufei Ma, Jiahao Li, Bin Li, and Yan Lu. Uncertainty-Aware Deep Video Compression with Ensembles. *arXiv preprint arXiv:2403.19158*, 2024. URL <https://arxiv.org/abs/2403.19158>.
- [163] Emilia Magnani, Marvin Pfortner, Tobias Weber, and Philipp Hennig. Linearization Turns Neural Operators into Function-Valued Gaussian Processes. *arXiv preprint arXiv:2406.05072*, 2024. URL <https://arxiv.org/abs/2406.05072>.
- [164] Ankur Mallick, Chaitanya Dwivedi, Bhavya Kailkhura, Gauri Joshi, and T. Yong-Jin Han.

- Probabilistic Neighbourhood Component Analysis: Sample Efficient Uncertainty Estimation in Deep Learning. *arXiv preprint arXiv:2007.10800*, 2020. URL <https://arxiv.org/abs/2007.10800>.
- [165] Sriram Mandalika, Lalitha V, and Athira Nambiar. PRIMEDrive-CoT: A Precognitive Chain-of-Thought Framework for Uncertainty-Aware Object Interaction in Driving Scene Scenario. *arXiv preprint arXiv:2504.05908*, 2025. URL <https://arxiv.org/abs/2504.05908>.
- [166] Gianluca Manzo and Julia Ive. Hesitation is defeat? Connecting Linguistic and Predictive Uncertainty. *arXiv preprint arXiv:2505.03910*, 2025. URL <https://arxiv.org/abs/2505.03910>.
- [167] Juan Maronas, Roberto Paredes, and Daniel Ramos. Calibration of Deep Probabilistic Models with Decoupled Bayesian Neural Networks. *arXiv preprint arXiv:1908.08972*, 2019. URL <https://arxiv.org/abs/1908.08972>.
- [168] Patrick McClure and Nikolaus Kriegeskorte. Robustly representing uncertainty in deep neural networks through sampling. *arXiv preprint arXiv:1611.01639*, 2016. URL <https://arxiv.org/abs/1611.01639>.
- [169] Xuhui Meng. Variational inference in neural functional prior using normalizing flows: Application to differential equation and operator learning problems. *arXiv preprint arXiv:2302.10448*, 2023. URL <https://arxiv.org/abs/2302.10448>.
- [170] Aaron Mishkin, Frederik Kunstner, Didrik Nielsen, Mark Schmidt, and Mohammad Emtiyaz Khan. SLANG: Fast Structured Covariance Approximations for Bayesian Deep Learning with Natural Gradient. *arXiv preprint arXiv:1811.04504*, 2018. URL <https://arxiv.org/abs/1811.04504>.
- [171] John Mitros and Brian Mac Namee. On the Validity of Bayesian Neural Networks for Uncertainty Estimation. *arXiv preprint arXiv:1912.01530*, 2019. URL <https://arxiv.org/abs/1912.01530>.
- [172] John Mitros, Arjun Pakrashi, and Brian Mac Namee. Ramifications of Approximate Posterior Inference for Bayesian Deep Learning in Adversarial and Out-of-Distribution Settings. *arXiv preprint arXiv:2009.01798*, 2020. URL <https://arxiv.org/abs/2009.01798>.
- [173] Devina Mohan and Anna Scaife. MCMC to address model misspecification in Deep Learning classification of Radio Galaxies. *arXiv preprint arXiv:2311.08243*, 2023. URL <https://arxiv.org/abs/2311.08243>.
- [174] Devina Mohan and Anna M. M. Scaife. Natural gradient descent for improving variational inference based classification of radio galaxies. *arXiv preprint arXiv:2511.13224*, 2025. URL <https://arxiv.org/abs/2511.13224>.
- [175] Melodie Monod, Alessandro Micheli, and Samir Bhatt. NeuralSurv: Deep Survival Analysis with Bayesian Uncertainty Quantification. *arXiv preprint arXiv:2505.11054*, 2025. URL <https://arxiv.org/abs/2505.11054>.
- [176] Hyeonbin Moon, Hanbin Cho, Wabi Demeke, Byungki Ryu, and Seunghwa Ryu. Thermal Conductivity Estimation of Thermoelectric Materials with Uncertainty Quantification Using Bayesian Physics-Informed Neural Networks. *arXiv preprint arXiv:2510.16723*, 2025. URL <https://arxiv.org/abs/2510.16723>.
- [177] Masaki Morimoto, Kai Fukami, Romit Maulik, Ricardo Vinuesa, and Koji Fukagata. Assessments of epistemic uncertainty using Gaussian stochastic weight averaging for fluid-flow regression. *arXiv preprint arXiv:2109.08248*, 2021. URL <https://arxiv.org/abs/2109.08248>.
- [178] Zahra Moslemi, Yang Meng, Shiwei Lan, and Babak Shahbaba. Scaling Up Bayesian Neural Networks with Neural Networks. *arXiv preprint arXiv:2312.11799*, 2023. URL <https://arxiv.org/abs/2312.11799>.
- [179] Lukas Mosser and Ehsan Zabihi Naeini. Calibration and Uncertainty Quantification of Bayesian Convolutional Neural Networks for Geophysical Applications. *arXiv preprint arXiv:2105.12115*, 2021. URL <https://arxiv.org/abs/2105.12115>.
- [180] Michel Moukari, Loic Simon, Sylvaine Picard, and Frederic Jurie. n-MeRCI: A new Metric to Evaluate the Correlation Between Predictive Uncertainty and True Error. *arXiv preprint arXiv:1908.07253*, 2019. URL <https://arxiv.org/abs/1908.07253>.
- [181] Hanieh Mousavi and Jeff D. Eldredge. Low-Order Flow Reconstruction and Uncertainty Quantification in Disturbed Aerodynamics Using Sparse Pressure Measurements. *arXiv preprint arXiv:2501.03406*, 2025. URL <https://arxiv.org/abs/2501.03406>.
- [182] Jishnu Mukhoti and Yarin Gal. Evaluating Bayesian Deep Learning Methods for Semantic Segmentation. *arXiv preprint arXiv:1811.12709*, 2018. URL <https://arxiv.org/abs/1811.12709>.
- [183] Adam T. Muller, Tobias Rogelein, and Nicolaj C. Stache. Monte Carlo Stochastic Depth for Uncertainty Estimation in Deep Learning. *arXiv preprint arXiv:2604.12719*, 2026. URL <https://arxiv.org/abs/2604.12719>.
- [184] Lorenzo Mur-Labadia and Ruben Martinez-Cantin. Bayesian deep learning of affordances from RGB images. *arXiv preprint arXiv:2109.12845*, 2021. URL <https://arxiv.org/abs/2109.12845>.

- [185] Abdulmajid Murad, Frank Alexander Kraemer, Kerstin Bach, and Gavin Taylor. Probabilistic Deep Learning to Quantify Uncertainty in Air Quality Forecasting. *arXiv preprint arXiv:2112.02622*, 2021. URL <https://arxiv.org/abs/2112.02622>.
- [186] Matthew Ng, Fumin Guo, Labonny Biswas, Stefan E. Petersen, Stefan K. Piechnik, Stefan Neubauer, and Graham Wright. Estimating Uncertainty in Neural Networks for Cardiac MRI Segmentation: A Benchmark Study. *arXiv preprint arXiv:2012.15772*, 2020. URL <https://arxiv.org/abs/2012.15772>.
- [187] Andre T. Nguyen, Fred Lu, Gary Lopez Munoz, Edward Raff, Charles Nicholas, and James Holt. Out of Distribution Data Detection Using Dropout Bayesian Neural Networks. *arXiv preprint arXiv:2202.08985*, 2022. URL <https://arxiv.org/abs/2202.08985>.
- [188] Dan Nguyen, Azar Sadeghnejad Barkousaraie, Gyanendra Bohara, Anjali Balagopal, Rafe McBeth, Mu-Han Lin, and Steve Jiang. A comparison of Monte Carlo dropout and bootstrap aggregation on the performance and uncertainty estimation in radiation therapy dose prediction with deep learning neural networks. *arXiv preprint arXiv:2011.00388*, 2020. URL <https://arxiv.org/abs/2011.00388>.
- [189] Sebastian W. Ober. Towards Improved Variational Inference for Deep Bayesian Models. *arXiv preprint arXiv:2401.12418*, 2024. URL <https://arxiv.org/abs/2401.12418>.
- [190] Rafael Orozco, Ali Siahkoobi, Mathias Louboutin, and Felix J. Herrmann. ASPIRE: Iterative Amortized Posterior Inference for Bayesian Inverse Problems. *arXiv preprint arXiv:2405.05398*, 2024. URL <https://arxiv.org/abs/2405.05398>.
- [191] Luis A. Ortega, Simon Rodriguez Santana, and Daniel Hernandez-Lobato. Variational Linearized Laplace Approximation for Bayesian Deep Learning. *arXiv preprint arXiv:2302.12565*, 2023. URL <https://arxiv.org/abs/2302.12565>.
- [192] Luis A. Ortega, Simon Rodriguez-Santana, and Daniel Hernandez-Lobato. Scalable Linearized Laplace Approximation via Surrogate Neural Kernel. *arXiv preprint arXiv:2601.21835*, 2026. URL <https://arxiv.org/abs/2601.21835>.
- [193] Jangseop Park and Namwoo Kang. Bayesian Mesh Optimization for Graph Neural Networks to Enhance Engineering Performance Prediction. *arXiv preprint arXiv:2406.01996*, 2024. URL <https://arxiv.org/abs/2406.01996>.
- [194] George D. Pasparakis, Lori Graham-Brady, and Michael D. Shields. Bayesian neural networks for predicting uncertainty in full-field material response. *arXiv preprint arXiv:2406.14838*, 2024. URL <https://arxiv.org/abs/2406.14838>.
- [195] Lukas Peron, Paolo Calafiura, Xiangyang Ju, and Jay Chan. Uncertainty Quantification and Propagation for ACORN, a geometric deep learning tracking pipeline for HEP experiments. *arXiv preprint arXiv:2508.16518*, 2025. URL <https://arxiv.org/abs/2508.16518>.
- [196] Pranav Poduval, Hrushikesh Loya, and Amit Sethi. Functional Space Variational Inference for Uncertainty Estimation in Computer Aided Diagnosis. *arXiv preprint arXiv:2005.11797*, 2020. URL <https://arxiv.org/abs/2005.11797>.
- [197] Jary Pomponi, Simone Scardapane, and Aurelio Uncini. Bayesian Neural Networks With Maximum Mean Discrepancy Regularization. *arXiv preprint arXiv:2003.00952*, 2020. URL <https://arxiv.org/abs/2003.00952>.
- [198] Augustin Prado, Ravinath Kausik, and Lalitha Venkataramanan. Dual Neural Network Architecture for Determining Epistemic and Aleatoric Uncertainties. *arXiv preprint arXiv:1910.06153*, 2019. URL <https://arxiv.org/abs/1910.06153>.
- [199] Kevin Raina. Logit Disagreement: OoD Detection with Bayesian Neural Networks. *arXiv preprint arXiv:2502.15648*, 2025. URL <https://arxiv.org/abs/2502.15648>.
- [200] Vineeth Rakesh and Swayambhoo Jain. Efficacy of Bayesian Neural Networks in Active Learning. *arXiv preprint arXiv:2104.00896*, 2021. URL <https://arxiv.org/abs/2104.00896>.
- [201] Omini Rathore, Richard Paul, Abigail Morrison, Hanno Scharr, and Elisabeth Pfahler. Efficient Epistemic Uncertainty Estimation in Cerebrovascular Segmentation. *arXiv preprint arXiv:2503.22271*, 2025. URL <https://arxiv.org/abs/2503.22271>.
- [202] Rahul D Ray and Utkarsh Srivastava. VOLTA: The Surprising Ineffectiveness of Auxiliary Losses for Calibrated Deep Learning. *arXiv preprint arXiv:2604.08639*, 2026. URL <https://arxiv.org/abs/2604.08639>.
- [203] Alfredo Reichlin, Miguel Vasco, and Danica Kragic. Walking on the Fiber: A Simple Geometric Approximation for Bayesian Neural Networks. *arXiv preprint arXiv:2512.01500*, 2025. URL <https://arxiv.org/abs/2512.01500>.
- [204] Daniel Ries, Jason Adams, and Joshua Zollweg. Target Detection on Hyperspectral Images Using MCMC and VI Trained Bayesian Neural Networks. *arXiv preprint arXiv:2308.06293*, 2023. URL <https://arxiv.org/abs/2308.06293>.

- [205] Lorenzo Rimella and Nick Whiteley. Hidden Markov Neural Networks. *arXiv preprint arXiv:2004.06963*, 2020. URL <https://arxiv.org/abs/2004.06963>.
- [206] Hippolyt Ritter, Martin Kukla, Cheng Zhang, and Yingzhen Li. Sparse Uncertainty Representation in Deep Learning with Inducing Weights. *arXiv preprint arXiv:2105.14594*, 2021. URL <https://arxiv.org/abs/2105.14594>.
- [207] Tommy Rochussen. Amortised Inference in Bayesian Neural Networks. *arXiv preprint arXiv:2309.03018*, 2023. URL <https://arxiv.org/abs/2309.03018>.
- [208] Johanna Rock, Tiago Azevedo, Rene de Jong, Daniel Ruiz-Munoz, and Partha Maji. On Efficient Uncertainty Estimation for Resource-Constrained Mobile Applications. *arXiv preprint arXiv:2111.09838*, 2021. URL <https://arxiv.org/abs/2111.09838>.
- [209] Julian Rodemann, Alexander Marquard, Thomas Augustin, and Michele Caprio. Self-Supervised Laplace Approximation for Bayesian Uncertainty Quantification. *arXiv preprint arXiv:2605.12208*, 2026. URL <https://arxiv.org/abs/2605.12208>.
- [210] Javier Rodriguez-Puigvert. Uncertainty and Self-Supervision in Single-View Depth. *arXiv preprint arXiv:2406.14226*, 2024. URL <https://arxiv.org/abs/2406.14226>.
- [211] Javier Rodriguez-Puigvert, Ruben Martinez-Cantin, and Javier Civera. Bayesian Deep Neural Networks for Supervised Learning of Single-View Depth. *arXiv preprint arXiv:2104.14202*, 2021. URL <https://arxiv.org/abs/2104.14202>.
- [212] Hrittik Roy, Marco Miani, Carl Henrik Ek, Philipp Hennig, Marvin Pfortner, Lukas Tatzel, and Sren Hauberg. Reparameterization invariance in approximate Bayesian inference. *arXiv preprint arXiv:2406.03334*, 2024. URL <https://arxiv.org/abs/2406.03334>.
- [213] Tim G. J. Rudner, Zonghao Chen, Yee Whye Teh, and Yarin Gal. Tractable Function-Space Variational Inference in Bayesian Neural Networks. *arXiv preprint arXiv:2312.17199*, 2023. URL <https://arxiv.org/abs/2312.17199>.
- [214] Sanne Ruijs, Alina Kosiakova, and Farrukh Javed. Beyond Accuracy: Reliability and Uncertainty Estimation in Convolutional Neural Networks. *arXiv preprint arXiv:2603.10731*, 2026. URL <https://arxiv.org/abs/2603.10731>.
- [215] Seongok Ryu and Sumin Lee. Accurate, reliable and interpretable solubility prediction of druglike molecules with attention pooling and Bayesian learning. *arXiv preprint arXiv:2210.07145*, 2022. URL <https://arxiv.org/abs/2210.07145>.
- [216] Seongok Ryu, Yongchan Kwon, and Woo Youn Kim. Uncertainty quantification of molecular property prediction using Bayesian neural network models. *arXiv preprint arXiv:1905.06945*, 2018. URL <https://arxiv.org/abs/1905.06945>.
- [217] Abhinav Sagar. Uncertainty Quantification using Variational Inference for Biomedical Image Segmentation. *arXiv preprint arXiv:2008.07588*, 2020. URL <https://arxiv.org/abs/2008.07588>.
- [218] Albert Saiapin and Kim Batselier. Laplace Approximation for Bayesian Tensor Network Kernel Machines. *arXiv preprint arXiv:2604.26673*, 2026. URL <https://arxiv.org/abs/2604.26673>.
- [219] Dylan Sam, Rattana Pukdee, Daniel P. Jeong, Yewon Byun, and J. Zico Kolter. Bayesian Neural Networks with Domain Knowledge Priors. *arXiv preprint arXiv:2402.13410*, 2024. URL <https://arxiv.org/abs/2402.13410>.
- [220] Colin Samplawski, Adam D. Cobb, Manoj Acharya, Ramneet Kaur, and Susmit Jha. Scalable Bayesian Low-Rank Adaptation of Large Language Models via Stochastic Variational Subspace Inference. *arXiv preprint arXiv:2506.21408*, 2025. URL <https://arxiv.org/abs/2506.21408>.
- [221] Paul Scemama and Ariel Kapusta. On the Out-of-Distribution Coverage of Combining Split Conformal Prediction and Bayesian Deep Learning. *arXiv preprint arXiv:2311.12688*, 2023. URL <https://arxiv.org/abs/2311.12688>.
- [222] Jannik Schmitt and Stefan Roth. Sampling-free Variational Inference for Neural Networks with Multiplicative Activation Noise. *arXiv preprint arXiv:2103.08497*, 2021. URL <https://arxiv.org/abs/2103.08497>.
- [223] David J. Schodt, Ryan Brown, Michael Merritt, Samuel Park, Delsin Menolascino, and Mark A. Peot. A Framework for Variational Inference of Lightweight Bayesian Neural Networks with Heteroscedastic Uncertainties. *arXiv preprint arXiv:2402.14532*, 2024. URL <https://arxiv.org/abs/2402.14532>.
- [224] John S. Schreck, David John Gagne, Charlie Becker, William E. Chapman, Kim Elmore, Da Fan, Gabrielle Gantos, Eliot Kim, Dhamma Kimpara, Thomas Martin, Maria J. Molina, Vanessa M. Pryzbylo, Jacob Radford, Belen Saavedra, Justin Willson, and Christopher Wirz. Evidential Deep Learning: Enhancing Predictive Uncertainty Estimation for Earth System Science Applications. *arXiv preprint arXiv:2309.13207*, 2023. URL <https://arxiv.org/abs/2309.13207>.
- [225] Kajetan Schweighofer, Lukas Aichberger, Mykyta Ielanskyi, Gunter Klambauer, and Sepp Hochreiter. Quantification of Uncertainty with Adversar-

- ial Models. *arXiv preprint arXiv:2307.03217*, 2023. URL <https://arxiv.org/abs/2307.03217>.
- [226] Nabeel Seedat and Christopher Kanan. Towards calibrated and scalable uncertainty representations for neural networks. *arXiv preprint arXiv:1911.00104*, 2019. URL <https://arxiv.org/abs/1911.00104>.
- [227] Florian Seligmann, Philipp Becker, Michael Volpp, and Gerhard Neumann. Beyond Deep Ensembles: A Large-Scale Evaluation of Bayesian Deep Learning under Distribution Shift. *arXiv preprint arXiv:2306.12306*, 2023. URL <https://arxiv.org/abs/2306.12306>.
- [228] Babak Shahbaba and Zahra Moslemi. Neural-Inspired Posterior Approximation (NIPA). *arXiv preprint arXiv:2601.22539*, 2026. URL <https://arxiv.org/abs/2601.22539>.
- [229] Alisa Sheinkman and Sara Wade. Understanding the Trade-offs in Accuracy and Uncertainty Quantification: Architecture and Inference Choices in Bayesian Neural Networks. *arXiv preprint arXiv:2503.11808*, 2025. URL <https://arxiv.org/abs/2503.11808>.
- [230] Che-Yung Shen, Xilin Yang, Yuzhu Li, Leon Lenk, and Aydogan Ozcan. Automated HER2 scoring with uncertainty quantification using lensfree holography and deep learning. *arXiv preprint arXiv:2601.18219*, 2026. URL <https://arxiv.org/abs/2601.18219>.
- [231] Maohao Shen, J. Jon Ryu, Soumya Ghosh, Yuheng Bu, Prasanna Sattigeri, Subhro Das, and Gregory W. Wornell. Are Uncertainty Quantification Capabilities of Evidential Deep Learning a Mirage? *arXiv preprint arXiv:2402.06160*, 2024. URL <https://arxiv.org/abs/2402.06160>.
- [232] Kumar Shridhar, Felix Laumann, and Marcus Liwicki. Uncertainty Estimations by Softplus normalization in Bayesian Convolutional Neural Networks with Variational Inference. *arXiv preprint arXiv:1806.05978*, 2018. URL <https://arxiv.org/abs/1806.05978>.
- [233] Kumar Shridhar, Felix Laumann, and Marcus Liwicki. A Comprehensive guide to Bayesian Convolutional Neural Network with Variational Inference. *arXiv preprint arXiv:1901.02731*, 2019. URL <https://arxiv.org/abs/1901.02731>.
- [234] Rajeev Kumar Singh, Rohan Gorantla, Sai Giridhar Allada, and Narra Pratap. SkiNet: A Deep Learning Solution for Skin Lesion Diagnosis with Uncertainty Estimation and Explainability. *arXiv preprint arXiv:2012.15049*, 2020. URL <https://arxiv.org/abs/2012.15049>.
- [235] Riccardo Di Sipio. Toward Ethical AI Through Bayesian Uncertainty in Neural Question Answering. *arXiv preprint arXiv:2512.17677*, 2025. URL <https://arxiv.org/abs/2512.17677>.
- [236] Emanuel Sommer and David Rugamer. Position: The Time for Sampling Is Now! Charting a New Course for Bayesian Deep Learning. *arXiv preprint arXiv:2605.21765*, 2026. URL <https://arxiv.org/abs/2605.21765>.
- [237] Romeo Sommerfeld, Christian Helms, and Ralf Herbrich. Approximate Message Passing for Bayesian Neural Networks. *arXiv preprint arXiv:2501.15573*, 2025. URL <https://arxiv.org/abs/2501.15573>.
- [238] Akshay Bhagwan Sonawane, Lena D. Swamikanan, and Lakshman Tamil. Robust Classification of Oral Cancer with Limited Training Data. *arXiv preprint arXiv:2510.01547*, 2025. URL <https://arxiv.org/abs/2510.01547>.
- [239] Anika Tabassum and Naimul Khan. An Efficient Confidence Measure-Based Evaluation Metric for Breast Cancer Screening Using Bayesian Neural Networks. *arXiv preprint arXiv:2008.05566*, 2020. URL <https://arxiv.org/abs/2008.05566>.
- [240] Jiawei Tang, Xinyan Du, Hui Liu, Junhui Hou, and Yuheng Jia. Variational Inference for Evidential Deep Learning. *arXiv preprint arXiv:2605.26477*, 2026. URL <https://arxiv.org/abs/2605.26477>.
- [241] Piotr Tempczyk, Ksawery Smoczynski, Philip Smolenski-Jensen, and Marek Cygan. One Simple Trick to Fix Your Bayesian Neural Network. *arXiv preprint arXiv:2207.13167*, 2022. URL <https://arxiv.org/abs/2207.13167>.
- [242] Claire Theobald, Frederic Pennerath, Brieuc Conan-Guez, Miguel Couceiro, and Amedeo Napoli. A Bayesian Neural Network based on Dropout Regulation. *arXiv preprint arXiv:2102.01968*, 2021. URL <https://arxiv.org/abs/2102.01968>.
- [243] Ponkrshnan Thiagarajan, Tamer A. Zaki, and Michael D. Shields. Accelerating Hamiltonian Monte Carlo for Bayesian Inference in Neural Networks and Neural Operators. *arXiv preprint arXiv:2507.14652*, 2025. URL <https://arxiv.org/abs/2507.14652>.
- [244] Katarina Tothova, Sarah Parisot, Matthew Lee, Esther Puyol-Anton, Andrew King, Marc Pollefeys, and Ender Konukoglu. Probabilistic 3D surface reconstruction from sparse MRI information. *arXiv preprint arXiv:2010.02041*, 2020. URL <https://arxiv.org/abs/2010.02041>.
- [245] Mame Diarra Toure and David A. Stephens. Not Just How Much, But Where: Decomposing Epistemic Uncertainty into Per-Class Contributions. *arXiv preprint arXiv:2602.21160*, 2026. URL <https://arxiv.org/abs/2602.21160>.

- [246] Mame Diarra Toure and David A. Stephens. Singular Bayesian Neural Networks. *arXiv preprint arXiv:2602.00387*, 2026. URL <https://arxiv.org/abs/2602.00387>.
- [247] Alex Treacher, Kevin Nguyen, Dylan Owens, Daniel Heitjan, and Albert Montillo. UQ-ARMED: Uncertainty quantification of adversarially-regularized mixed effects deep learning for clustered non-iid data. *arXiv preprint arXiv:2211.15888*, 2022. URL <https://arxiv.org/abs/2211.15888>.
- [248] Dennis Ulmer, Christian Hardmeier, and Jes Frellsen. Prior and Posterior Networks: A Survey on Evidential Deep Learning Methods For Uncertainty Estimation. *arXiv preprint arXiv:2110.03051*, 2021. URL <https://arxiv.org/abs/2110.03051>.
- [249] Ali Unlu and Laurence Aitchison. Variational Laplace for Bayesian neural networks. *arXiv preprint arXiv:2103.00222*, 2021. URL <https://arxiv.org/abs/2103.00222>.
- [250] Anshuk Uppal, Kristoffer Stensbo-Smidt, Wouter Boomsma, and Jes Frellsen. Implicit Variational Inference for High-Dimensional Posteriors. *arXiv preprint arXiv:2310.06643*, 2023. URL <https://arxiv.org/abs/2310.06643>.
- [251] Meet P. Vadera, Adam D. Cobb, Brian Jalaian, and Benjamin M. Marlin. Impact of Parameter Sparsity on Stochastic Gradient MCMC Methods for Bayesian Deep Learning. *arXiv preprint arXiv:2202.03770*, 2022. URL <https://arxiv.org/abs/2202.03770>.
- [252] Mahsa Valizadeh, Rui Tuo, and James Caverlee. BI-DCGAN: A Theoretically Grounded Bayesian Framework for Efficient and Diverse GANs. *arXiv preprint arXiv:2510.26892*, 2025. URL <https://arxiv.org/abs/2510.26892>.
- [253] Francisca Vasconcelos, Bobby He, Nalini Singh, and Yee Whye Teh. UncertaiNR: Uncertainty Quantification of End-to-End Implicit Neural Representations for Computed Tomography. *arXiv preprint arXiv:2202.10847*, 2022. URL <https://arxiv.org/abs/2202.10847>.
- [254] Daniel F. Villarraga and Ricardo A. Daziano. Bayesian Deep Learning for Discrete Choice. *arXiv preprint arXiv:2505.18077*, 2025. URL <https://arxiv.org/abs/2505.18077>.
- [255] Neng Wan, Dapeng Li, and Naira Hovakimyan. f-Divergence Variational Inference. *arXiv preprint arXiv:2009.13093*, 2020. URL <https://arxiv.org/abs/2009.13093>.
- [256] Lukasz Wandzik, Raul Vicente Garcia, and Jorg Kruger. Uncertainty Quantification in Deep Residual Neural Networks. *arXiv preprint arXiv:2007.04905*, 2020. URL <https://arxiv.org/abs/2007.04905>.
- [257] Bin Wang, Jie Lu, Zheng Yan, Huaishao Luo, Tianrui Li, Yu Zheng, and Guangquan Zhang. Deep Uncertainty Quantification: A Machine Learning Approach for Weather Forecasting. *arXiv preprint arXiv:1812.09467*, 2018. URL <https://arxiv.org/abs/1812.09467>.
- [258] Duo Wang, Yiren Zhao, Ilia Shumailov, and Robert Mullins. Model Architecture Adaption for Bayesian Neural Networks. *arXiv preprint arXiv:2202.04392*, 2022. URL <https://arxiv.org/abs/2202.04392>.
- [259] Kaizheng Wang, Keivan Shariatmadar, Shireen Kudukkil Manchingal, Fabio Cuzzolin, David Moens, and Hans Hallez. CreINNs: Credal-Set Interval Neural Networks for Uncertainty Estimation in Classification Tasks. *arXiv preprint arXiv:2401.05043*, 2024. URL <https://arxiv.org/abs/2401.05043>.
- [260] Kuan-Chieh Wang, Paul Vicol, James Lucas, Li Gu, Roger Grosse, and Richard Zemel. Adversarial Distillation of Bayesian Neural Network Posteriors. *arXiv preprint arXiv:1806.10317*, 2018. URL <https://arxiv.org/abs/1806.10317>.
- [261] Qi Wang and Herke van Hoof. Doubly Stochastic Variational Inference for Neural Processes with Hierarchical Latent Variables. *arXiv preprint arXiv:2008.09469*, 2020. URL <https://arxiv.org/abs/2008.09469>.
- [262] Zhongyao Wang, Taoyong Cui, Jiawen Zou, Shufei Zhang, Bo Yan, Wanli Ouyang, Weimin Tan, and Mao Su. Equivariant Evidential Deep Learning for Interatomic Potentials. *arXiv preprint arXiv:2602.10419*, 2026. URL <https://arxiv.org/abs/2602.10419>.
- [263] Tobias Jan Wiecek, Leon de Andrade, Thomas Mollenhoff, and Marcus Rohrbach. Calibrated Sampling-Free Uncertainty Estimation in Bayesian Deep Learning. *arXiv preprint arXiv:2606.16214*, 2026. URL <https://arxiv.org/abs/2606.16214>.
- [264] Veit D. Wild, Robert Hu, and Dino Sejdinovic. Generalized Variational Inference in Function Spaces: Gaussian Measures meet Bayesian Deep Learning. *arXiv preprint arXiv:2205.06342*, 2022. URL <https://arxiv.org/abs/2205.06342>.
- [265] Soohaeng Yoo Willow, Tae Hyeon Park, Gi Beom Sim, Sung Wook Moon, Seung Kyu Min, D. ChangMo Yang, Hyun Woo Kim, Juho Lee, and Chang Woo Myung. Bayesian E(3)-Equivariant Interatomic Potential with Iterative Restratification of Many-body Message Passing. *arXiv preprint arXiv:2510.03046*, 2025. URL <https://arxiv.org/abs/2510.03046>.

- [266] Jae Oh Woo. Analytic Mutual Information in Bayesian Neural Networks. *arXiv preprint arXiv:2201.09815*, 2022. URL <https://arxiv.org/abs/2201.09815>.
- [267] Anqi Wu, Sebastian Nowozin, Edward Meeds, Richard E. Turner, Jose Miguel Hernandez-Lobato, and Alexander L. Gaunt. Deterministic Variational Inference for Robust Bayesian Neural Networks. *arXiv preprint arXiv:1810.03958*, 2018. URL <https://arxiv.org/abs/1810.03958>.
- [268] Bingzhe Wu, Chaochao Chen, Shiwan Zhao, Cen Chen, Yuan Yao, Guangyu Sun, Li Wang, Xiaolu Zhang, and Jun Zhou. Characterizing Membership Privacy in Stochastic Gradient Langevin Dynamics. *arXiv preprint arXiv:1910.02249*, 2019. URL <https://arxiv.org/abs/1910.02249>.
- [269] Mengjing Wu, Junyu Xuan, and Jie Lu. Functional Stochastic Gradient MCMC for Bayesian Neural Networks. *arXiv preprint arXiv:2409.16632*, 2024. URL <https://arxiv.org/abs/2409.16632>.
- [270] Tailin Wu, Willie Neiswanger, Hongtao Zheng, Stefano Ermon, and Jure Leskovec. Uncertainty Quantification for Forward and Inverse Problems of PDEs via Latent Global Evolution. *arXiv preprint arXiv:2402.08383*, 2024. URL <https://arxiv.org/abs/2402.08383>.
- [271] Tim Z. Xiao, Weiyang Liu, and Robert Bamber. A Compact Representation for Bayesian Neural Networks By Removing Permutation Symmetry. *arXiv preprint arXiv:2401.00611*, 2023. URL <https://arxiv.org/abs/2401.00611>.
- [272] Audrey Xie, Elhoucine Elfatimi, Sambuddha Ghosal, and Pratik Shah. Deep Learning with Uncertainty Quantification for Predicting the Segmentation Dice Coefficient of Prostate Cancer Biopsy Images. *arXiv preprint arXiv:2109.00115*, 2021. URL <https://arxiv.org/abs/2109.00115>.
- [273] Xinlei Xiong, Wenbo Hu, Shuxun Zhou, Kaifeng Bi, Lingxi Xie, Ying Liu, Richang Hong, and Qi Tian. Bridging the Gap Between Bayesian Deep Learning and Ensemble Weather Forecasts. *arXiv preprint arXiv:2511.14218*, 2025. URL <https://arxiv.org/abs/2511.14218>.
- [274] Hua Xu, Julian D. Arias-Londono, and Juan I. Godino-Llorente. Confidence-Uncertainty Boundary Calibration for Bayesian Deep Learning in Medical Image Analysis. *arXiv preprint arXiv:2602.11973*, 2026. URL <https://arxiv.org/abs/2602.11973>.
- [275] Jian Xu, Delu Zeng, and John Paisley. Sparse Inducing Points in Deep Gaussian Processes: Enhancing Modeling with Denoising Diffusion Variational Inference. *arXiv preprint arXiv:2407.17033*, 2024. URL <https://arxiv.org/abs/2407.17033>.
- [276] Lu Xu, Tsai Hor Chan, Kwok Fai Lam, Lequan Yu, and Guosheng Yin. Variational Polya Tree. *arXiv preprint arXiv:2510.22651*, 2025. URL <https://arxiv.org/abs/2510.22651>.
- [277] Xinnuo Xu, Minyoung Kim, Royson Lee, Brais Martinez, and Timothy Hospedales. A Bayesian Approach to Data Point Selection. *arXiv preprint arXiv:2411.03768*, 2024. URL <https://arxiv.org/abs/2411.03768>.
- [278] Yujia Xue, Shiyi Cheng, Yunzhe Li, and Lei Tian. Reliable deep-learning-based phase imaging with uncertainty quantification. *arXiv preprint arXiv:1901.02038*, 2019. URL <https://arxiv.org/abs/1901.02038>.
- [279] Weichi Yao, Bianca Dumitrascu, Bryan R. Goldsmith, and Yixin Wang. Goal-Oriented Influence-Maximizing Data Acquisition for Learning and Optimization. *arXiv preprint arXiv:2602.19578*, 2026. URL <https://arxiv.org/abs/2602.19578>.
- [280] Tong Ye, Shijing Si, Jianzong Wang, Ning Cheng, Zhitao Li, and Jing Xiao. On the Calibration and Uncertainty with Polya-Gamma Augmentation for Dialog Retrieval Models. *arXiv preprint arXiv:2303.08606*, 2023. URL <https://arxiv.org/abs/2303.08606>.
- [281] Jiaxiang Yi and Miguel A. Bessa. Cooperative Variance Estimation and Bayesian Neural Networks for Disentangling Aleatoric and Epistemic Uncertainties. *arXiv preprint arXiv:2505.02743*, 2025. URL <https://arxiv.org/abs/2505.02743>.
- [282] Taeseong Yoon and Heeyoung Kim. Uncertainty Estimation by Flexible Evidential Deep Learning. *arXiv preprint arXiv:2510.18322*, 2025. URL <https://arxiv.org/abs/2510.18322>.
- [283] Jialin Yu, Laila Alrajhi, Anoushka Harit, Zhongtian Sun, Alexandra I. Cristea, and Lei Shi. Exploring Bayesian Deep Learning for Urgent Instructor Intervention Need in MOOC Forums. *arXiv preprint arXiv:2104.12643*, 2021. URL <https://arxiv.org/abs/2104.12643>.
- [284] Linlin Yu, Bowen Yang, Tianhao Wang, Kangshuo Li, and Feng Chen. Predictive Uncertainty Quantification for Bird’s Eye View Segmentation: A Benchmark and Novel Loss Function. *arXiv preprint arXiv:2405.20986*, 2024. URL <https://arxiv.org/abs/2405.20986>.
- [285] Linlin Yu, Kangshuo Li, Pritom Kumar Saha, Yifei Lou, and Feng Chen. Evidential Uncertainty Probes for Graph Neural Networks. *arXiv preprint arXiv:2503.08097*, 2025. URL <https://arxiv.org/abs/2503.08097>.
- [286] Filippo Zacchei, Paolo Conti, Attilio Alberto Frangi, and Andrea Manzoni. Multi-Fidelity Delayed Acceptance: hierarchical MCMC sampling

- for Bayesian inverse problems combining multiple solvers through deep neural networks. *arXiv preprint arXiv:2512.16430*, 2025. URL <https://arxiv.org/abs/2512.16430>.
- [287] Sheheryar Zaidi, Arber Zela, Thomas Elsken, Chris Holmes, Frank Hutter, and Yee Whye Teh. Neural Ensemble Search for Uncertainty Estimation and Dataset Shift. *arXiv preprint arXiv:2006.08573*, 2020. URL <https://arxiv.org/abs/2006.08573>.
- [288] Jiaming Zeng, Adam Lesnikowski, and Jose M. Alvarez. The Relevance of Bayesian Layer Positioning to Model Uncertainty in Deep Bayesian Active Learning. *arXiv preprint arXiv:1811.12535*, 2018. URL <https://arxiv.org/abs/1811.12535>.
- [289] Zhe Zeng and Guy Van den Broeck. Collapsed Inference for Bayesian Deep Learning. *arXiv preprint arXiv:2306.09686*, 2023. URL <https://arxiv.org/abs/2306.09686>.
- [290] Kilian Zepf, Selma Wanna, Marco Miani, Juston Moore, Jes Frellsen, Sren Hauberg, Frederik Warburg, and Aasa Feragen. Laplacian Segmentation Networks Improve Epistemic Uncertainty Quantification. *arXiv preprint arXiv:2303.13123*, 2023. URL <https://arxiv.org/abs/2303.13123>.
- [291] Qishi Zhan, Minxuan Hu, Guansu Wang, Jiaxin Liu, and Liang He. Unstable Rankings in Bayesian Deep Learning Evaluation. *arXiv preprint arXiv:2604.23102*, 2026. URL <https://arxiv.org/abs/2604.23102>.
- [292] Guodong Zhang, Shengyang Sun, David Duvenaud, and Roger Grosse. Noisy Natural Gradient as Variational Inference. *arXiv preprint arXiv:1712.02390*, 2017. URL <https://arxiv.org/abs/1712.02390>.
- [293] Qiyiwen Zhang, Zhiqi Bu, Kan Chen, and Qi Long. Differentially Private Bayesian Neural Networks on Accuracy, Privacy and Reliability. *arXiv preprint arXiv:2107.08461*, 2021. URL <https://arxiv.org/abs/2107.08461>.
- [294] Yao Zhang and Alpha A. Lee. Bayesian semi-supervised learning for uncertainty-calibrated prediction of molecular properties and active learning. *arXiv preprint arXiv:1902.00925*, 2019. URL <https://arxiv.org/abs/1902.00925>.
- [295] Chengmin Zhou, Ville Kyrki, Pasi Franti, and Laura Ruotsalainen. Combining Bayesian Inference and Reinforcement Learning for Agent Decision Making: A Review. *arXiv preprint arXiv:2505.07911*, 2025. URL <https://arxiv.org/abs/2505.07911>.
- [296] Hao Henry Zhou, Yunyang Xiong, and Vikas Singh. Building Bayesian Neural Networks with Blocks: On Structure, Interpretability and Uncertainty. *arXiv preprint arXiv:1806.03563*, 2018. URL <https://arxiv.org/abs/1806.03563>.
- [297] Taotao Zhou, Enrique Lopez Droguett, Ali Mosleh, and Felix T. S. Chan. An Uncertainty-Informed Framework for Trustworthy Fault Diagnosis in Safety-Critical Applications. *arXiv preprint arXiv:2111.00874*, 2021. URL <https://arxiv.org/abs/2111.00874>.
- [298] Albert Zhu, Simon Batzner, Albert Musaelian, and Boris Kozinsky. Fast Uncertainty Estimates in Deep Learning Interatomic Potentials. *arXiv preprint arXiv:2211.09866*, 2022. URL <https://arxiv.org/abs/2211.09866>.
- [299] Xiao-Yan Zhu, Heng-Jian Si-Tu, Hao Zhang, Wei Gao, Wen-Bin Lin, and Xiao-Hua Li. Bayesian neural network with autoencoder for model-based description of Fe-Fe -particle preformation factor. *arXiv preprint arXiv:2603.07976*, 2026. URL <https://arxiv.org/abs/2603.07976>.
- [300] Yinhao Zhu and Nicholas Zabararas. Bayesian Deep Convolutional Encoder-Decoder Networks for Surrogate Modeling and Uncertainty Quantification. *arXiv preprint arXiv:1801.06879*, 2018. URL <https://arxiv.org/abs/1801.06879>.